

063-0605-00L : Computational Structural Design 1

Computational Graphic Statics

Dr. Juney Lee & Dr. Lluís Enrique

Autumn Semester 2021

Week 4

Friday, October 15th

Hour 1

09:45 – 09:50

- Recap of Exercise 2

09:50 – 10:30

- **Lecture 3 : Computational Graphic Statics**

10:30 – 10:45

- Break

Hour 2

10:45 – 11:35

- **Tutorial 3 : Procedural graphic statics 2**

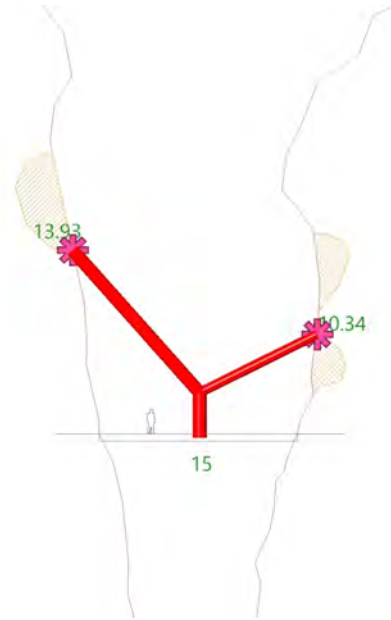
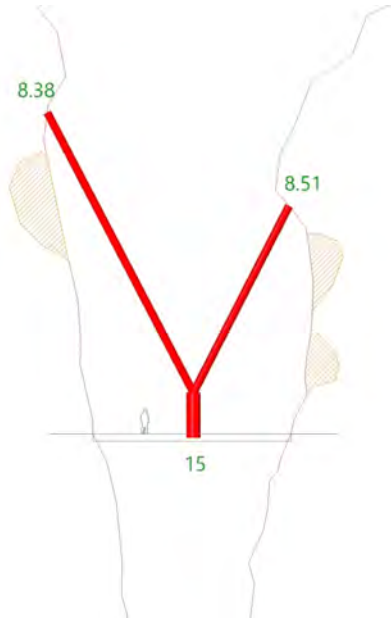
11:30 – 11:45

- Break

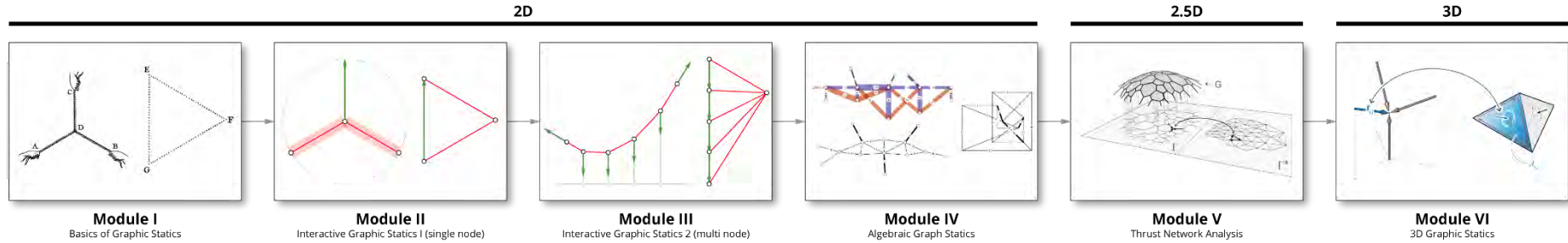
Hour 3

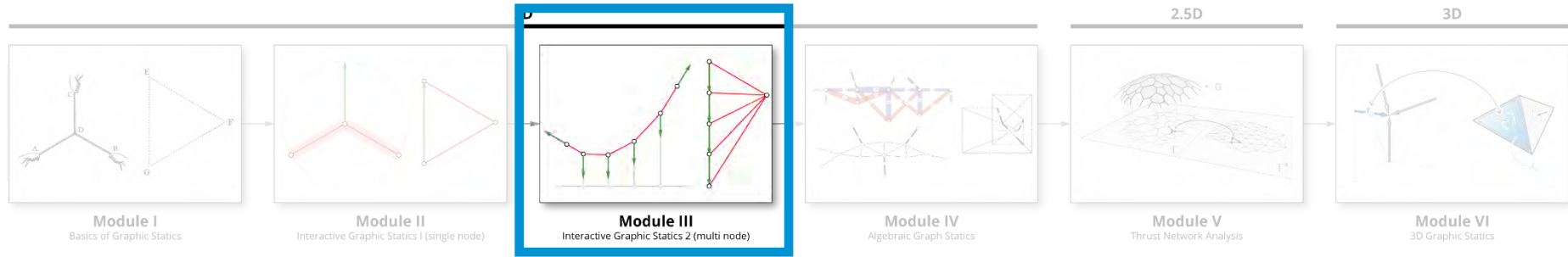
11:45 – 12:30

- **Tutorial 3 : Procedural graphic statics 2 (cont'd)**

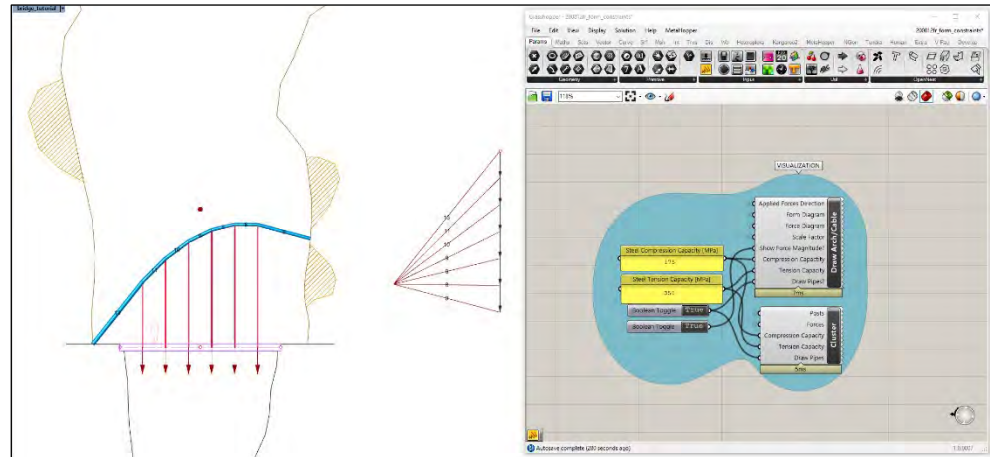
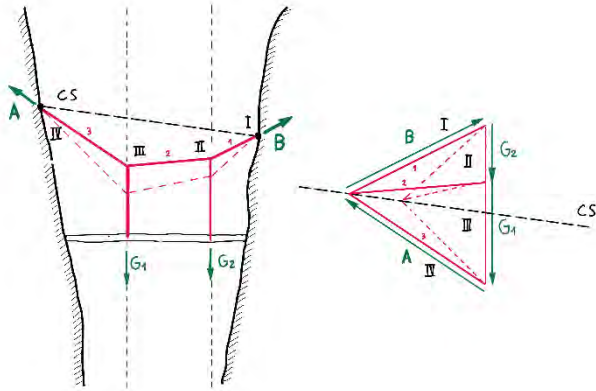


Structural Type(s)		Topic	Week(s)	Day	Topic	Instructor(s)
1. planar structures	cables & arches	Introduction	Week 1 (9/24)	1	Lecture 1	Dr. Juney Lee
				2	Tutorial 1	Dr. Lluís Enríque Lotte Aldinger
				3	Work session 1	<i>Exercise 1: Reciprocal diagrams in rhino + GH basics</i> Instructors
		2D GS (procedural)	Week 2 (10/1)	1	Lecture 2	Quick recap of Exercise 1 Algorithmic design & thinking Dr. Juney Lee
				2	Tutorial 2	Single node bridge in GH (form to force) Lotte Aldinger
				3		
	Week 3 (10/8)	1	Work session 2	<i>Exercise 2: Single node bridge with GH</i> Lotte Aldinger + Instructors		
		2				
		3				
	trusses	2D GS (AGS)	Week 4 (10/15)	1	Lecture 3	Quick recap of Exercise 2 Computational graphic statics Dr. Juney Lee
				2	Tutorial 3	Multi-node bridge in Grasshopper Dr. Lluís Enríque
				3		
Week 5 (10/22)		2				
		3				
		Week 6	seminar week			
2. surface structures	shells	2.5 GS (TNA)	Week 7 (11/5)	1	Lecture 4	Quick recap of Exercise 3 Algebraic graph statics (AGS) Dr. Juney Lee
				2	Tutorial 4	compas_ags + IGS (Interactive Graphic Statics) plugin Ricardo Avelino
				3		
		Week 8 (11/12)	1	Work session 4	<i>Exercise 4: Truss problems with IGS</i> Ricardo Avelino + Instructors	
			2			
			3			
3. spatial structures	polyhedral structures	3D GS (polyhedral)	Week 9 (11/19)	1	Lecture 5	Quick recap of Exercise 4 Thrust Network Analysis (TNA) Dr. Juney Lee
				2	Tutorial 5	compas_tna + RV2 (RhinoVAULT 2) plugin Dr. Juney Lee
				3		
		Week 10 (11/26)	1	Work session 5	<i>Exercise 5: Shell design exercises with RV2</i> Dr. Juney Lee + Instructors	
			2			
			3			
3. spatial structures	polyhedral structures	3D GS (polyhedral)	Week 11 (12/3)	1	Lecture 6	Quick recap of Exercise 5 3D graphic statics (3GS) Dr. Juney Lee
				2	Tutorial 6	compas_3gs + 3GS (3D graphic statics) plugin Dr. Juney Lee
				3	Work session 6	<i>Exercise 6: Spatial structures with 3GS</i>
		Week 12 (12/10)	1	Lecture 7	Quick recap of Exercise 6 Pt I. Outlook on computational graphic statics Dr. Juney Lee	
			2	Q/A	Pt II. Guest Lecture Discussions, feedback, evaluations, etc. Dr. Juney Lee TBD	





Module III · 2D procedural graphic statics 2



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Computational Graphic Statics

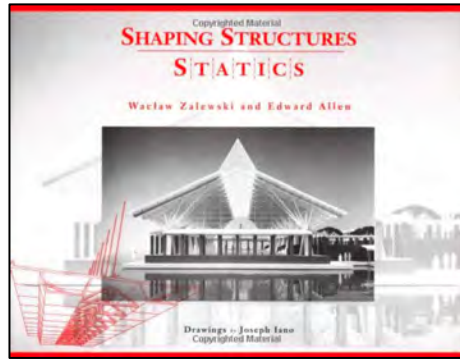
Lecture 3

Computational Graphic Statics

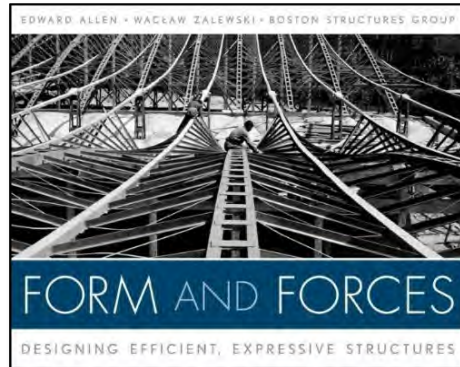
Friday, October 15th, 2021

Dr. Juney Lee

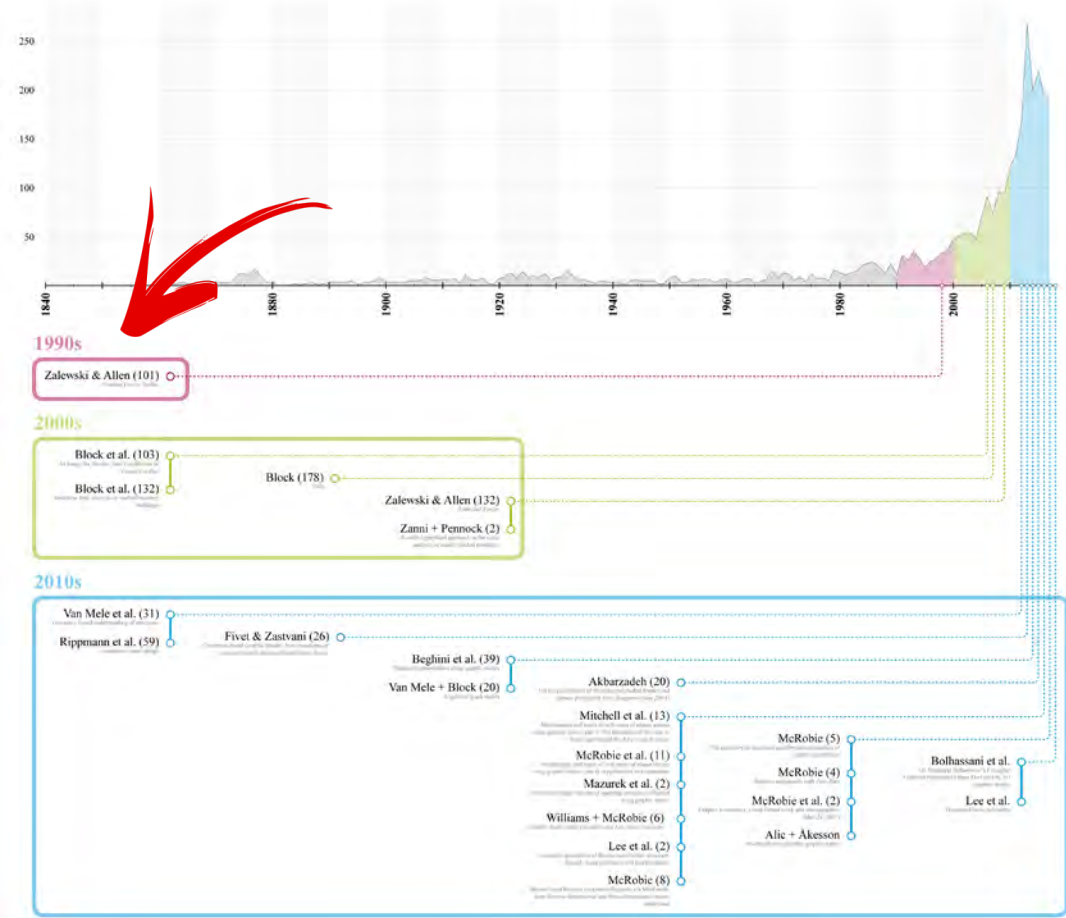
Graphic Statics

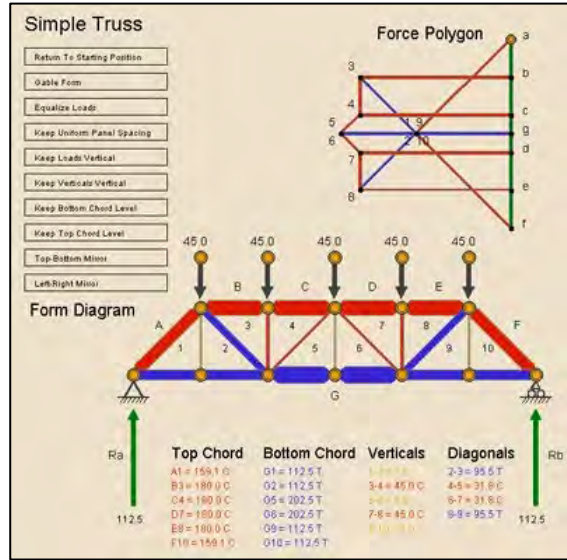


Shaping Structures
Zalewski and Allen (1997)

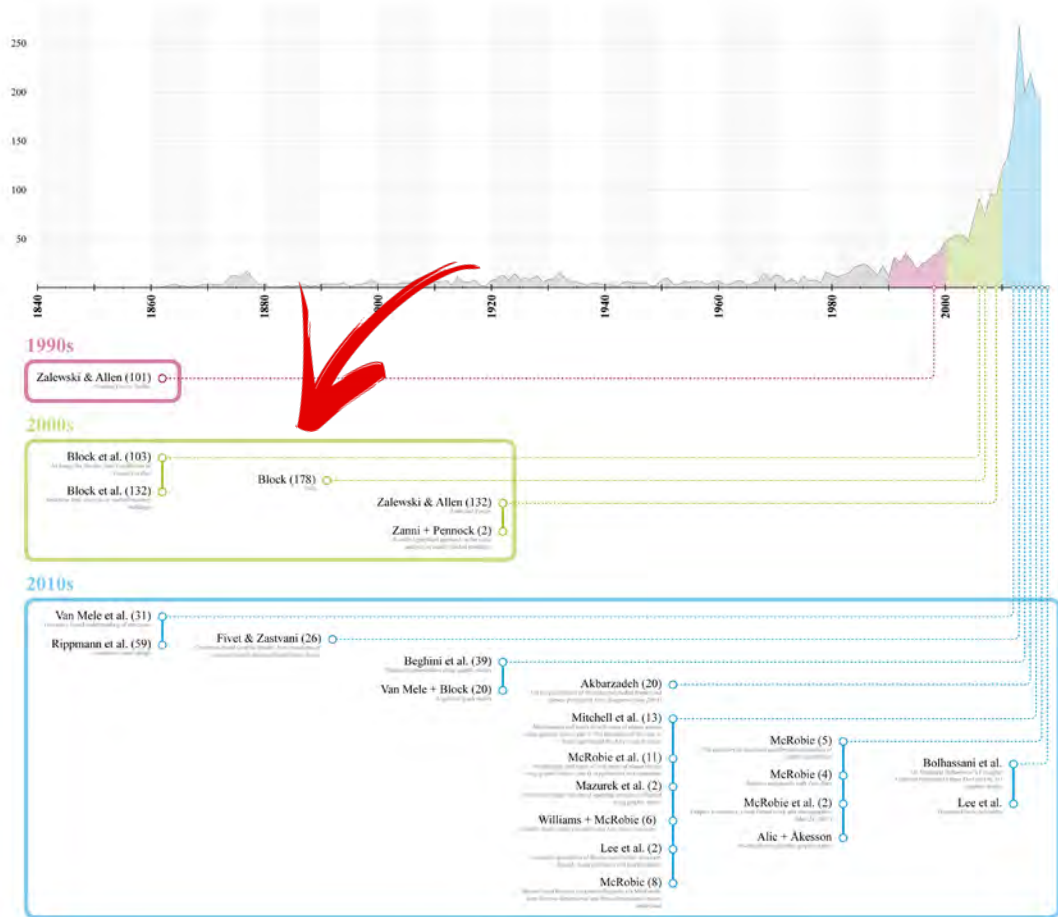


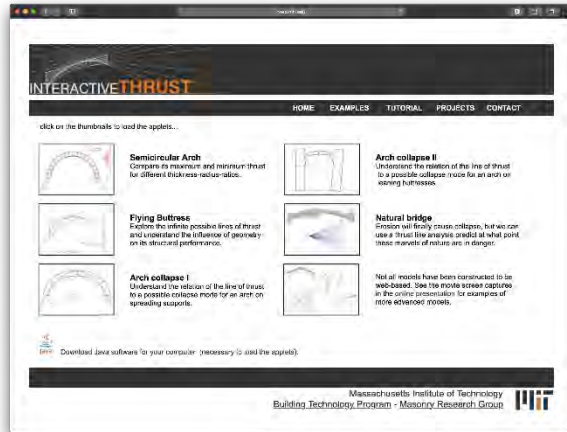
Form and Forces
(Allen and Zalewski (2010))



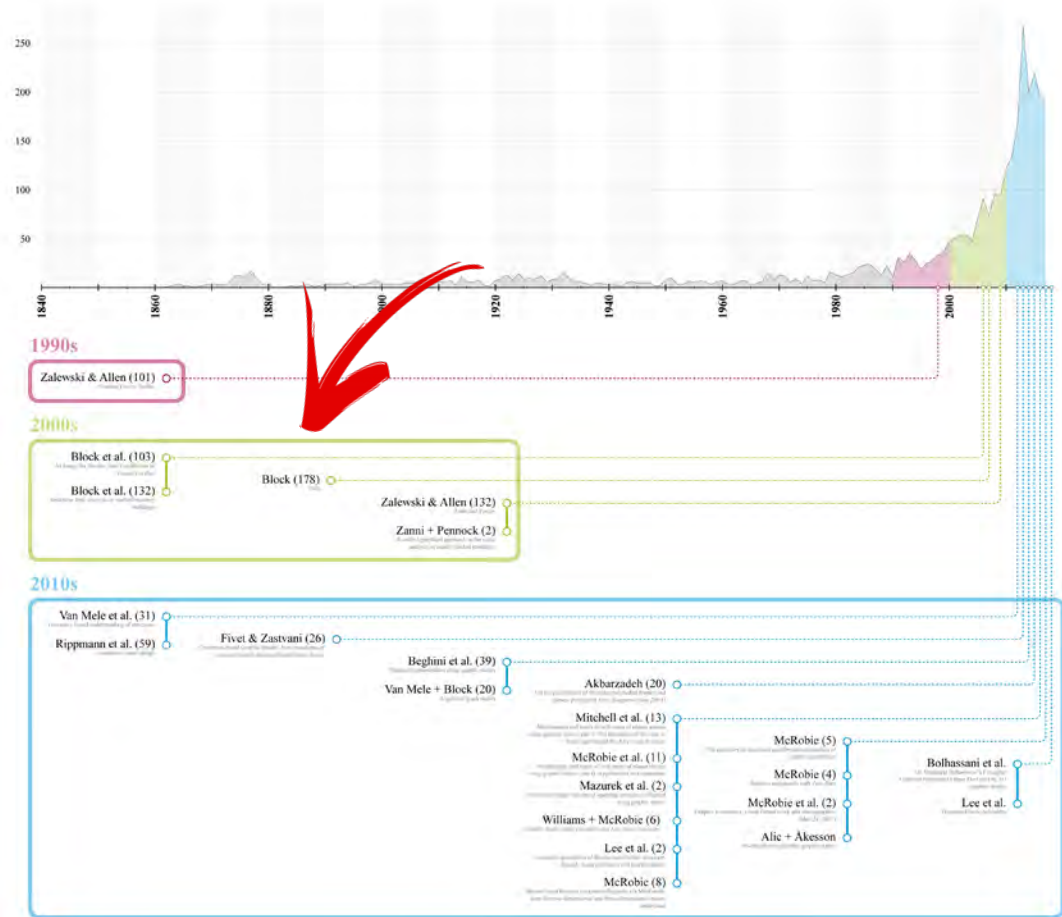


Active Statics
Greenwood and Allen (2003)





InteractiveTHRUST
Block and Ochsendorf (2005)





eQUILIBRIUM

an interactive environment for graphic statics-based structural design

my navigation ▼

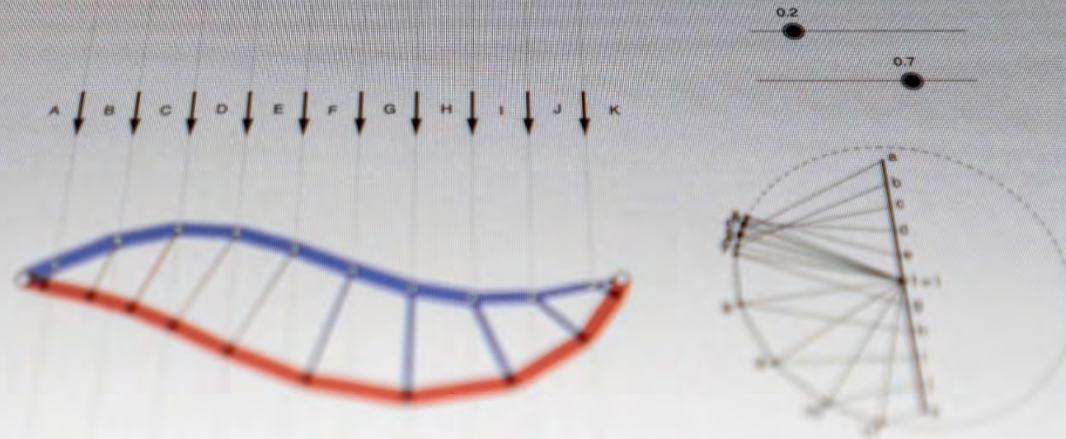
examples

case studies

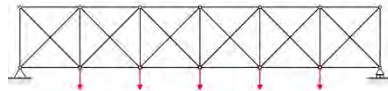
drawings

courses

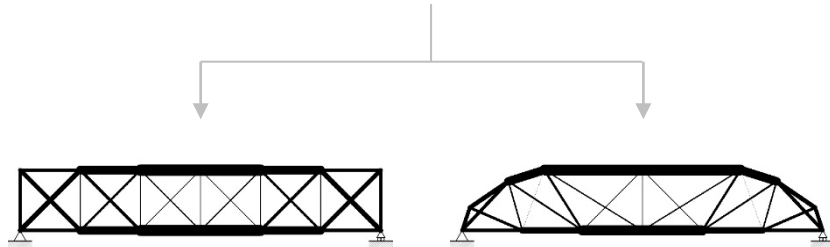
exams



Form analysis



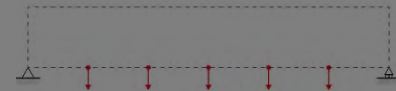
Pre-determined Shape



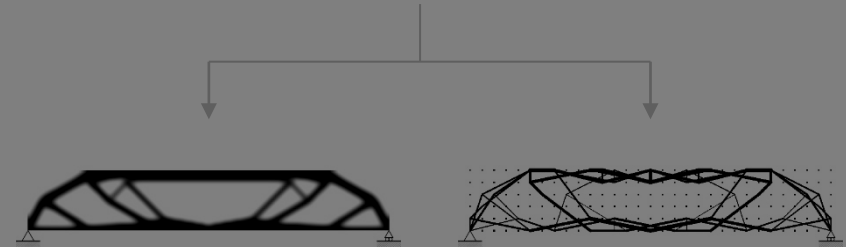
Size

Shape / Geometry

Form generation



Unknown Shape



Topology (Continuum)

Topology (Discrete)

Form analysis

Interactive (parametric)

- Analysis
- Constraint-driven design

Computational graphic statics

- Algebraic graph(ic) statics (AGS)
- Bi-directional AGS

Optimization

Form generation

Form finding

- Thrust network analysis (TNA)
- 3D graphic statics

Generative design

- Subdivision graphic statics
- Shape grammars

Visualisation

...

Form analysis

Interactive (parametric)

Analysis

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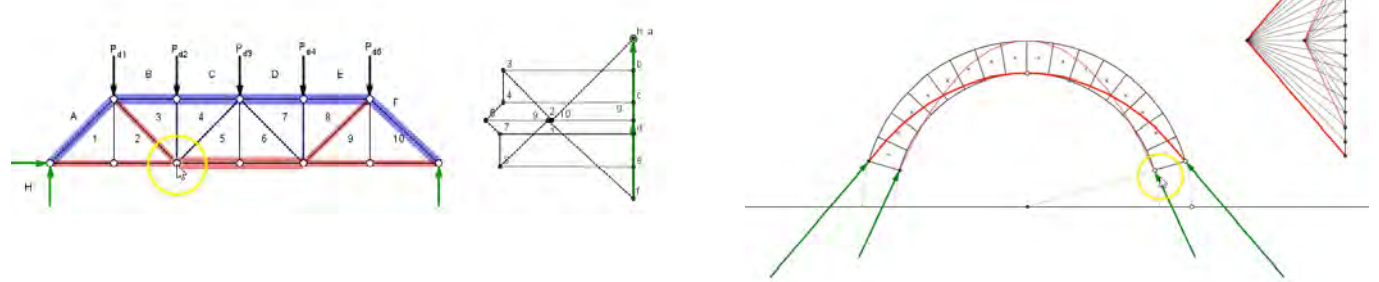
Generative design

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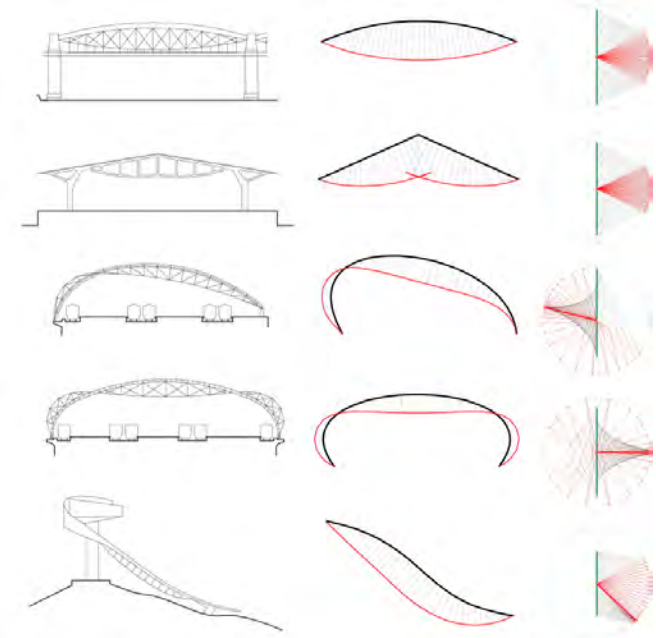
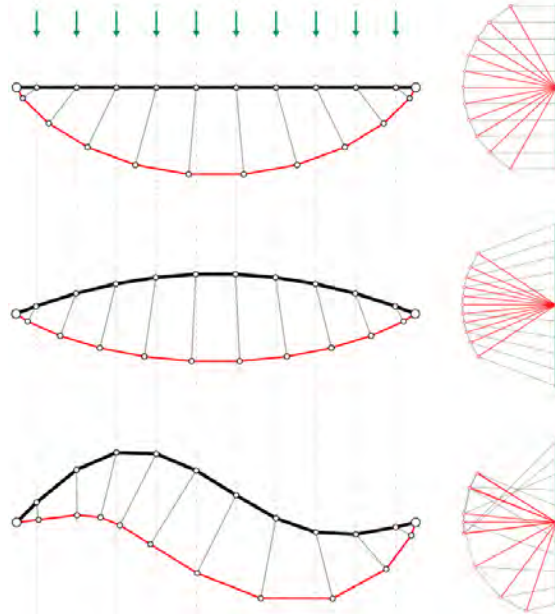
Generative design

Subdivision graphic statics

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Visualisation

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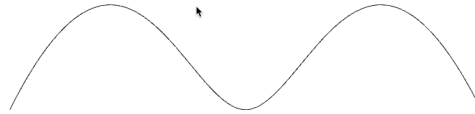
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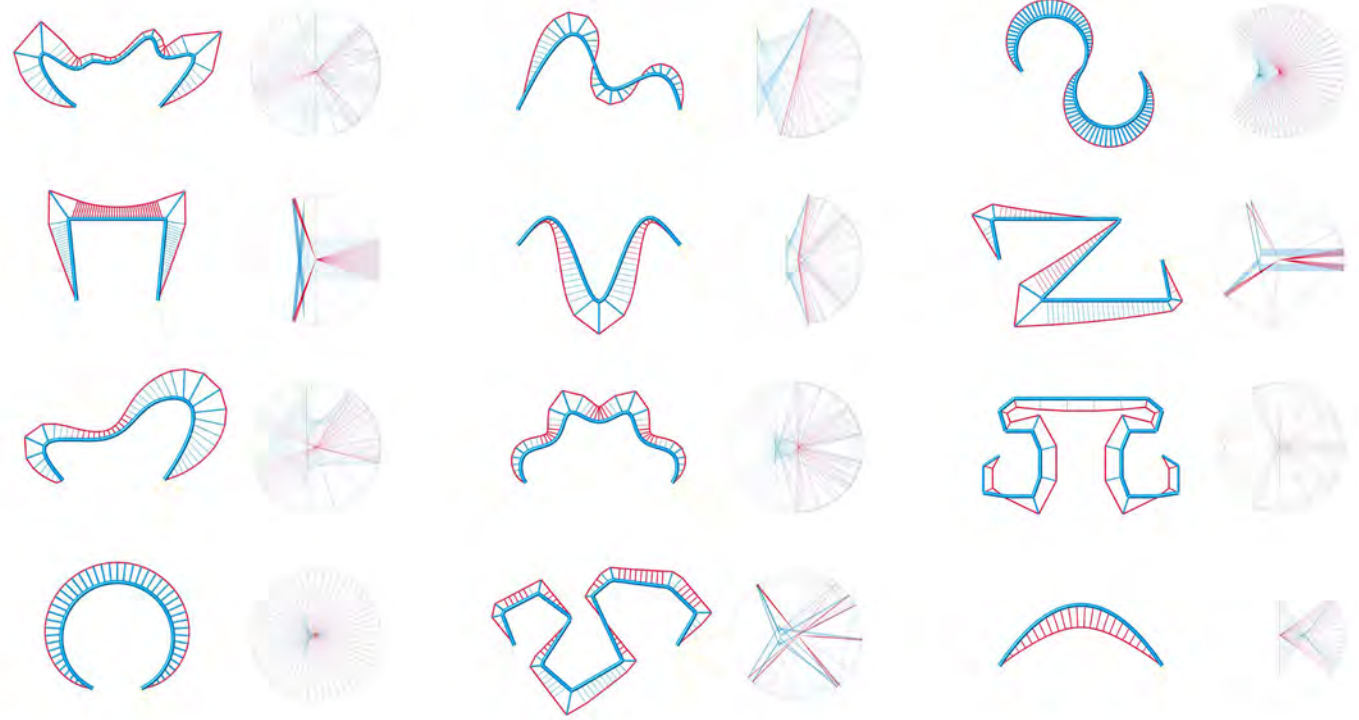
Generative design

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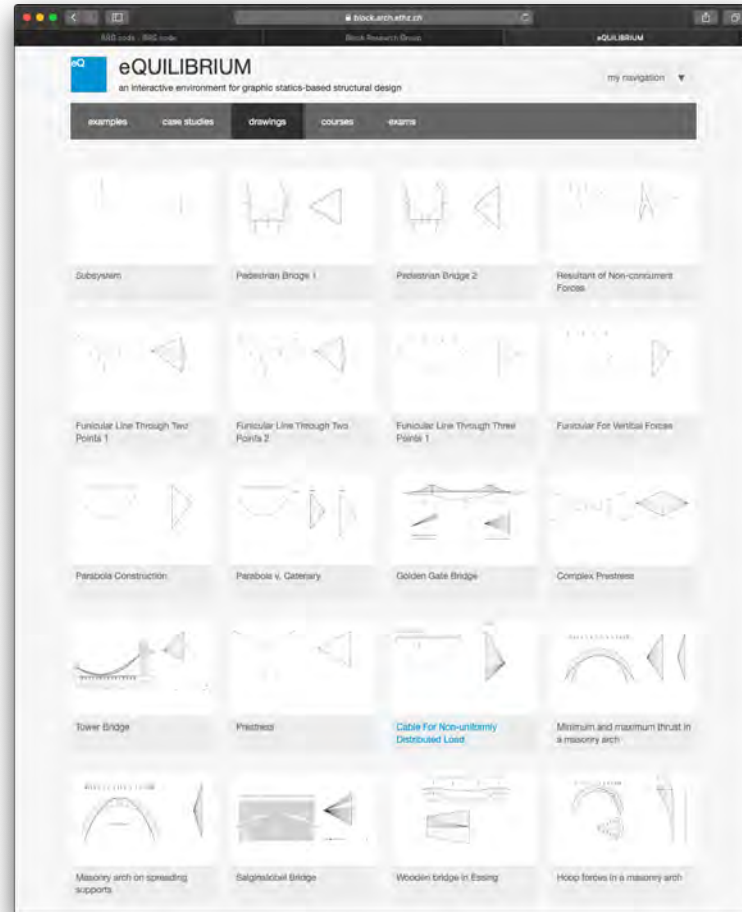
Generative design

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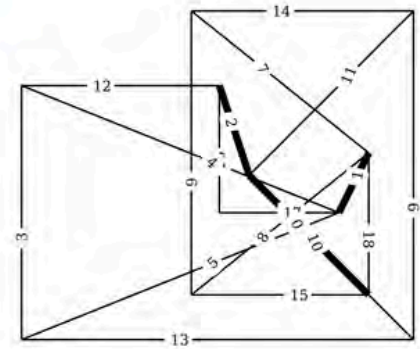
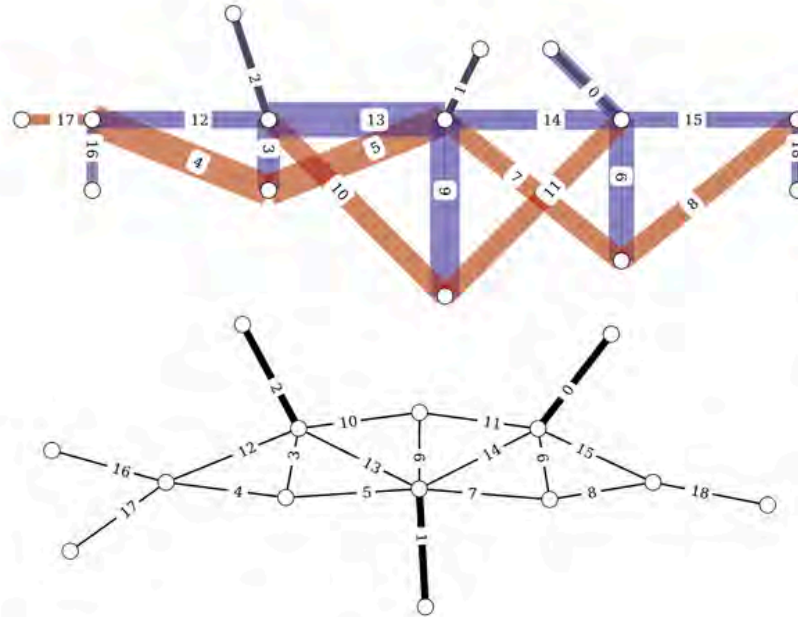
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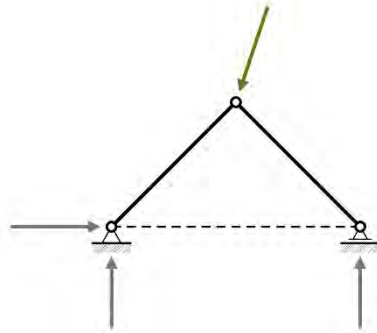
Thrust network analysis (TNA)
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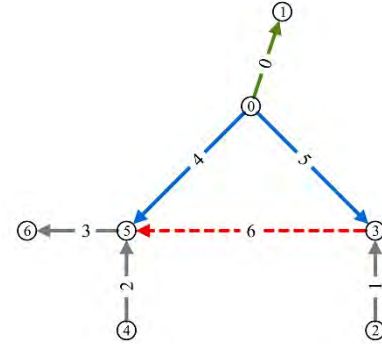
Subdivision graphic statics
Shape grammars

Visualisation

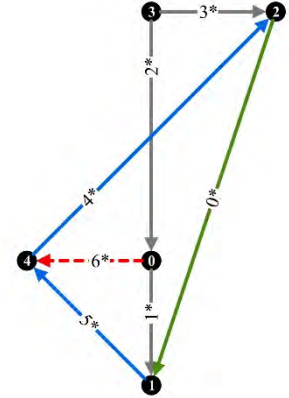
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Form diagram



Form diagram as a graph



Force diagram

Get topology and connectivity

Solve via equilibrium matrix

Form analysis

Interactive (parametric)

Analysis

Constraint-driven design

Computational graphic statics

Algebraic graph(ic) statics (AGS)

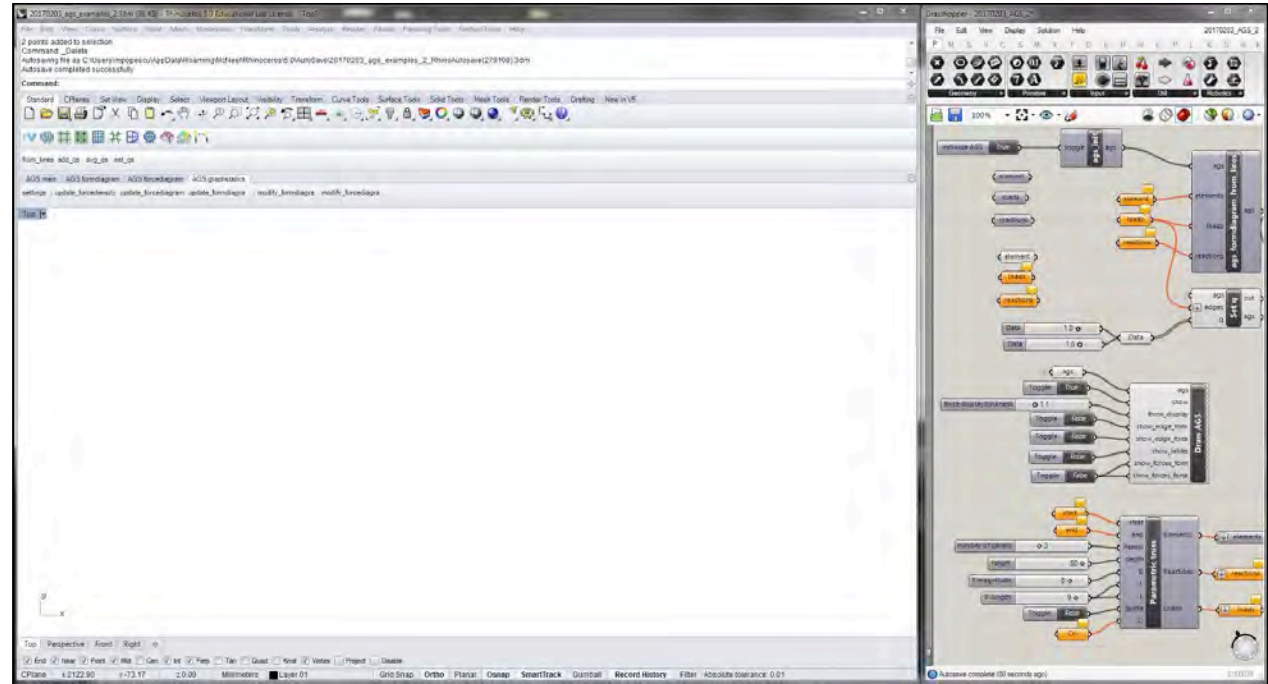
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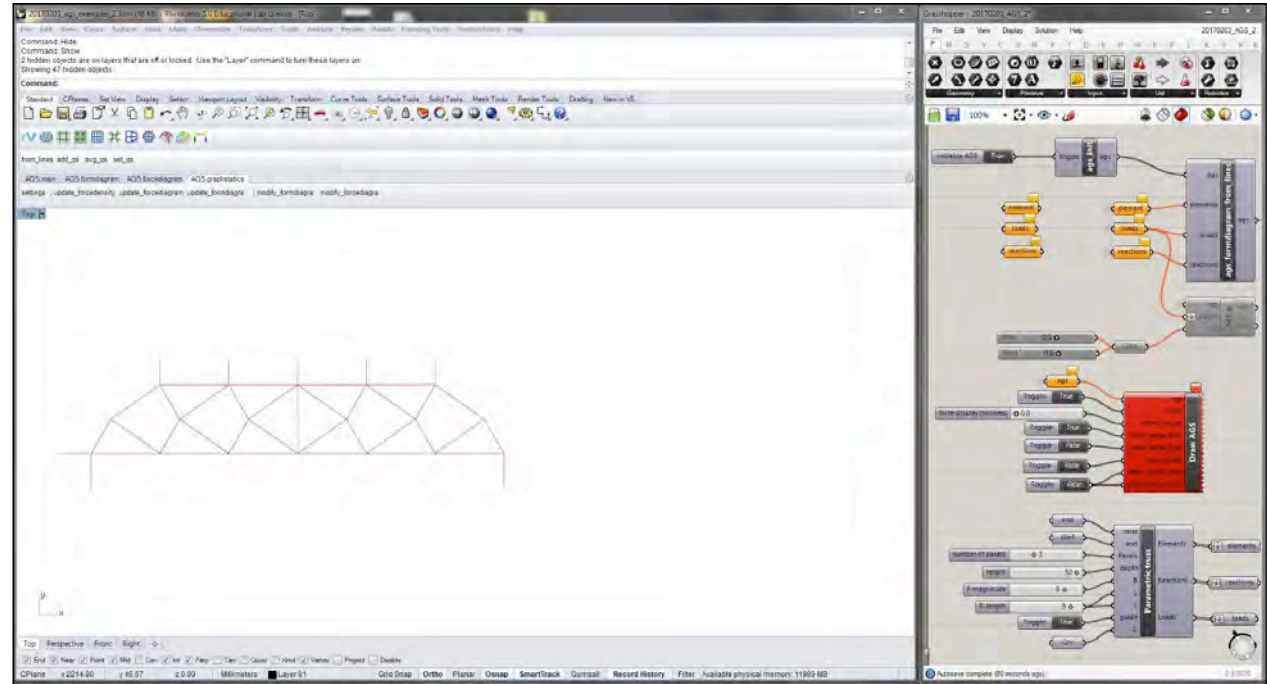
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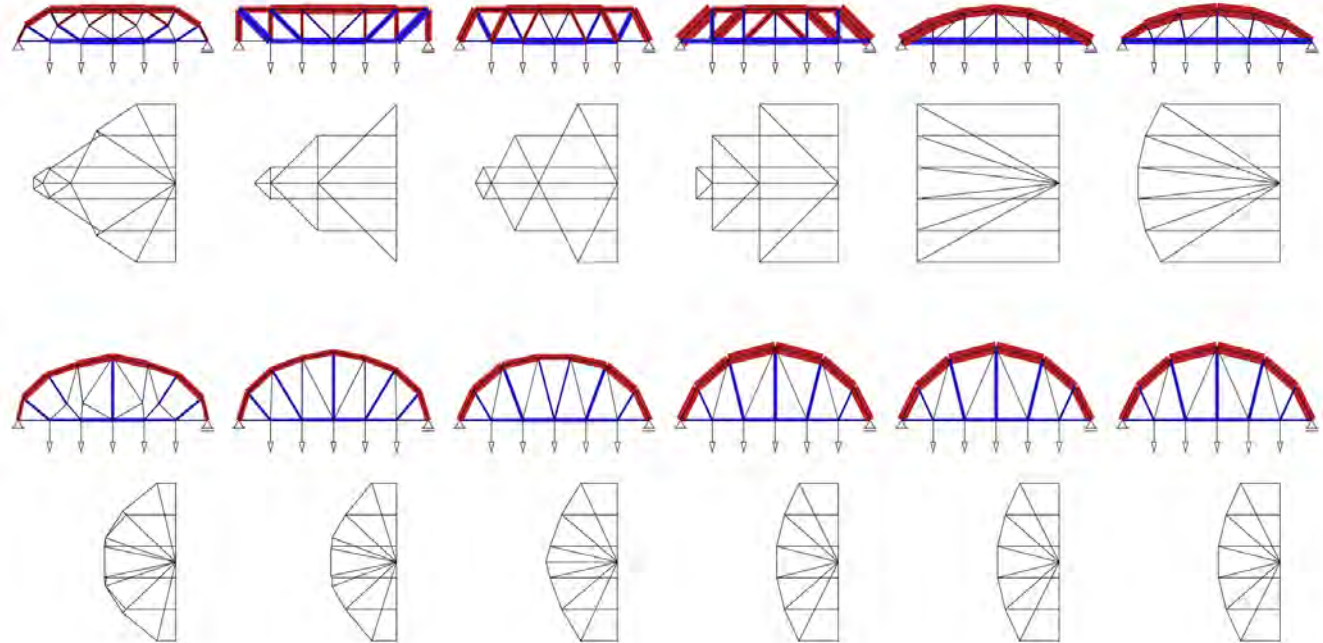
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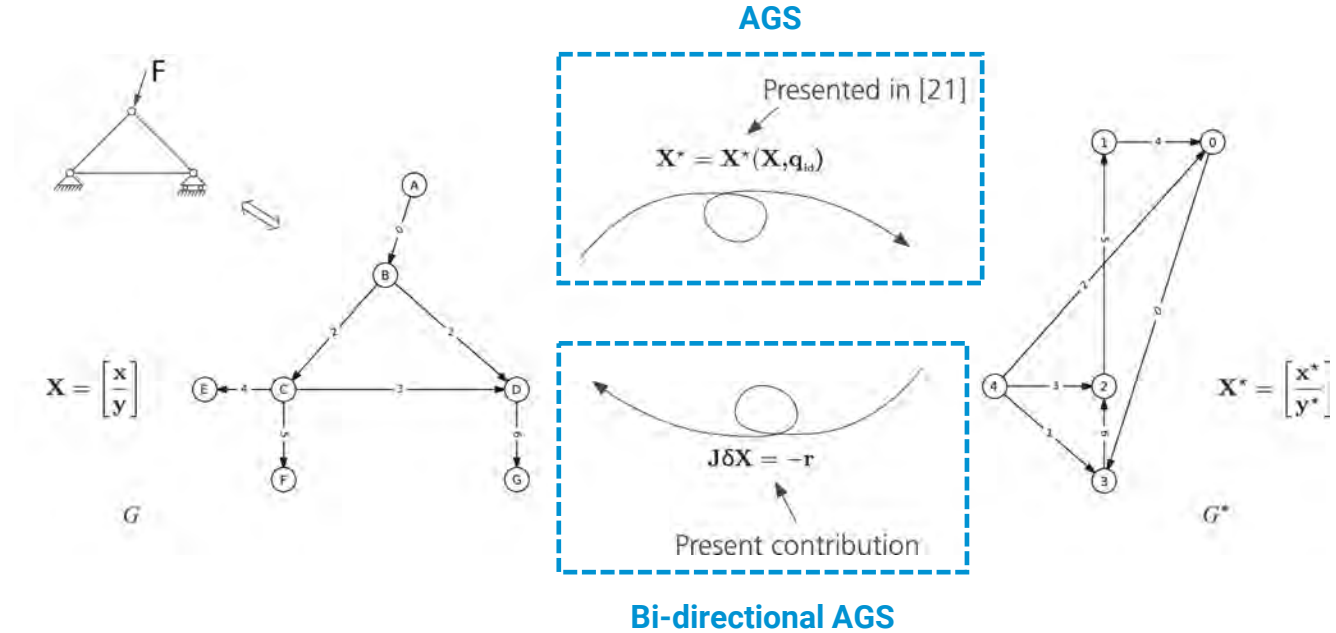
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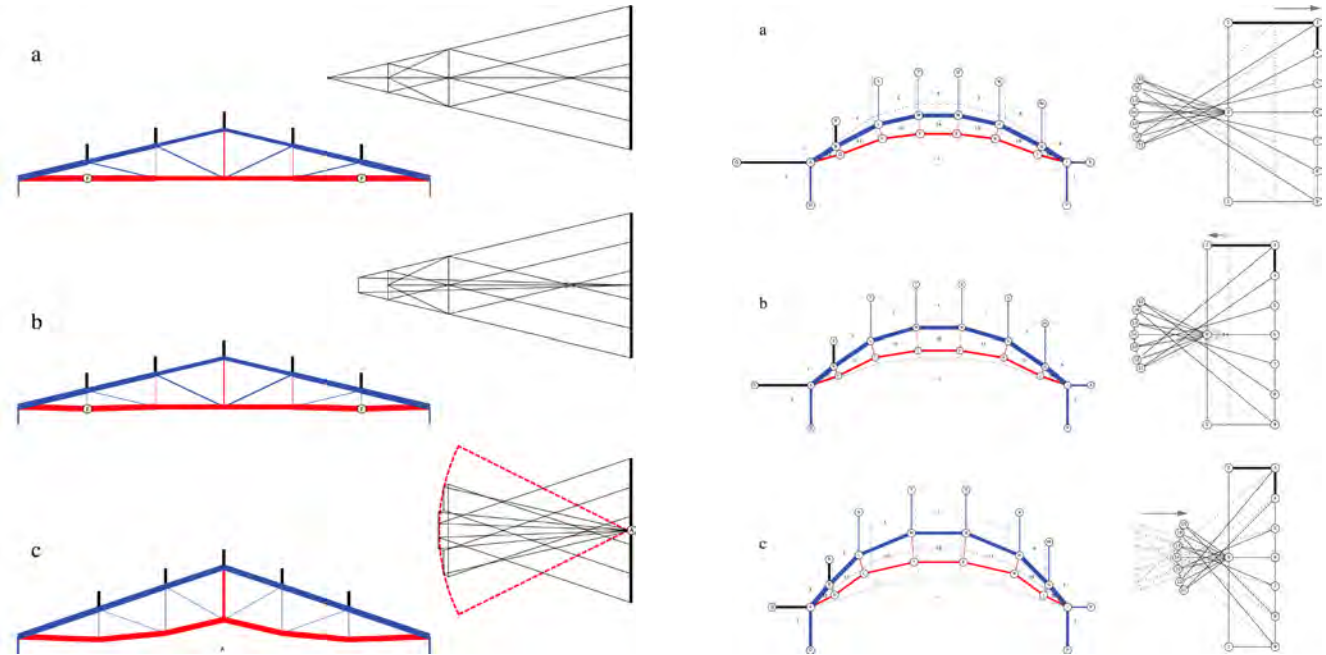
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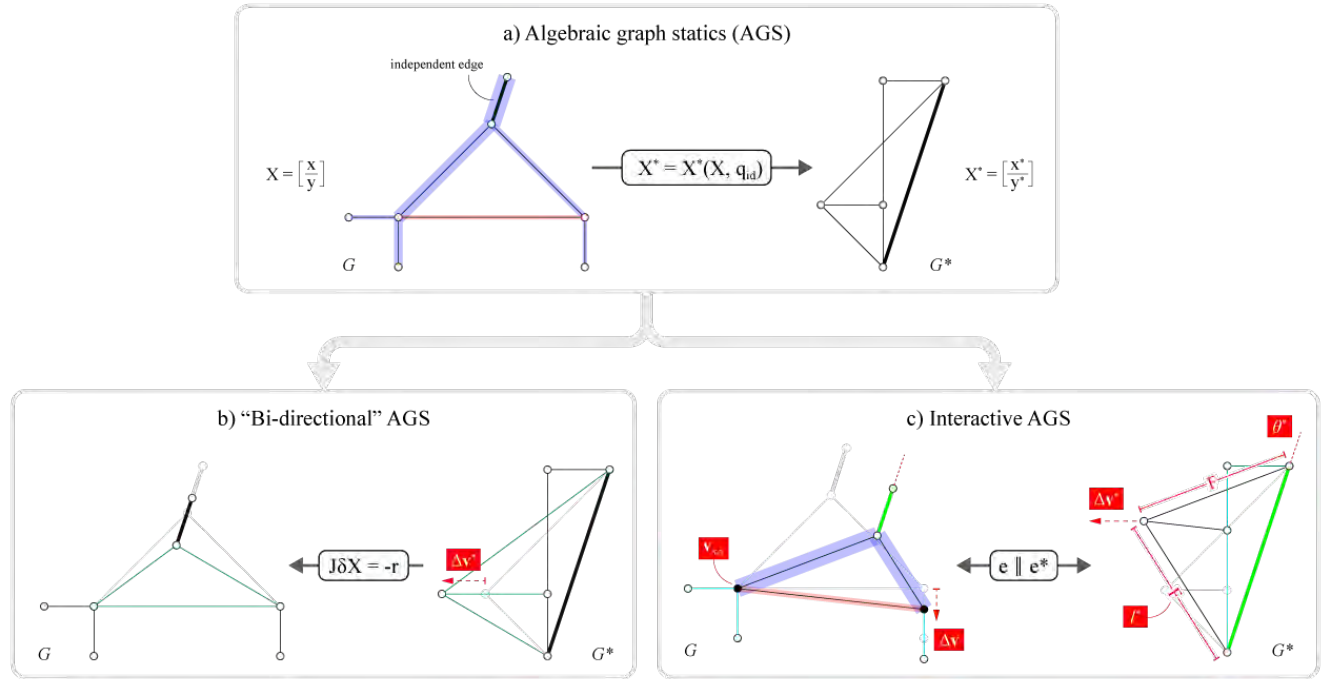
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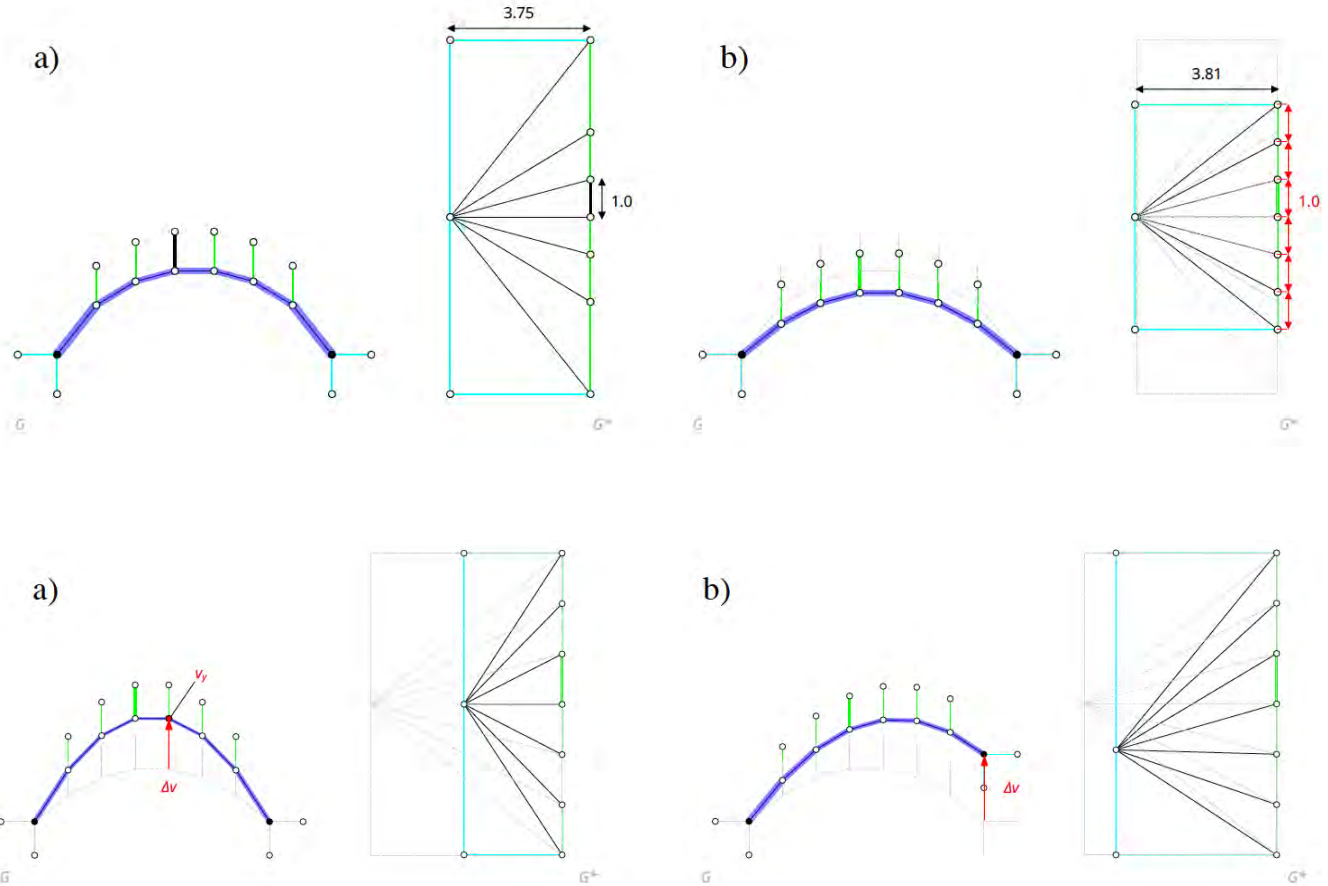
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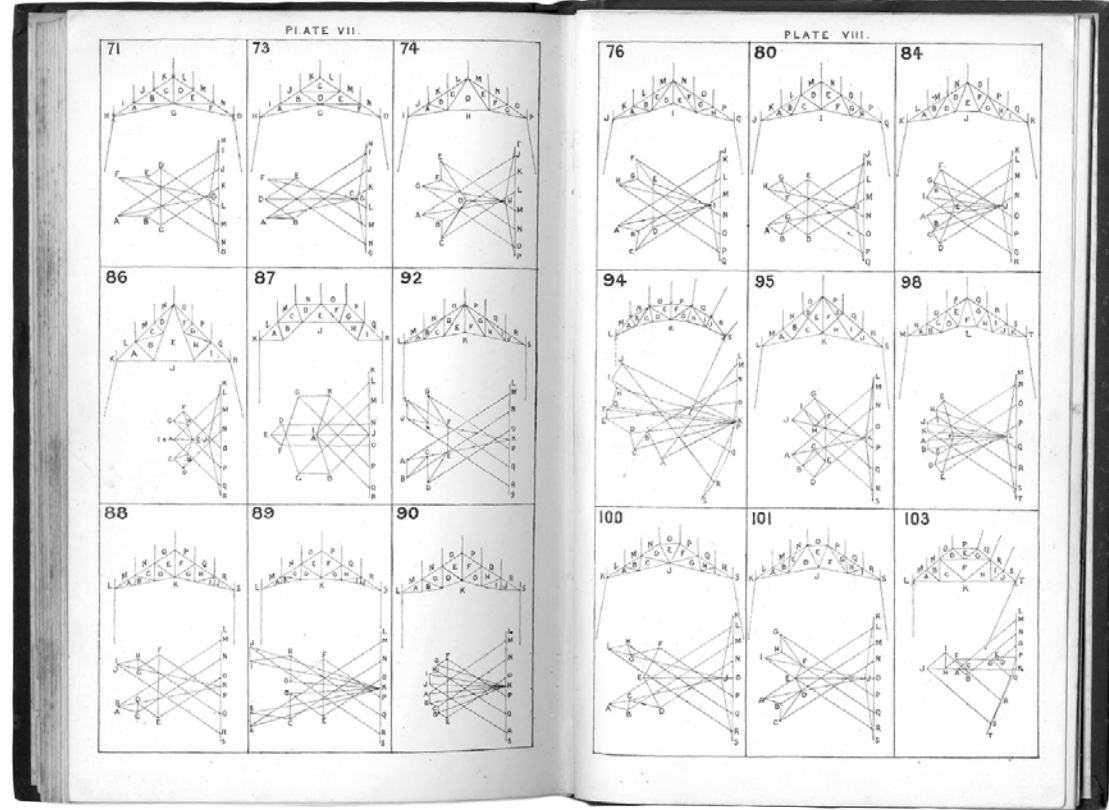
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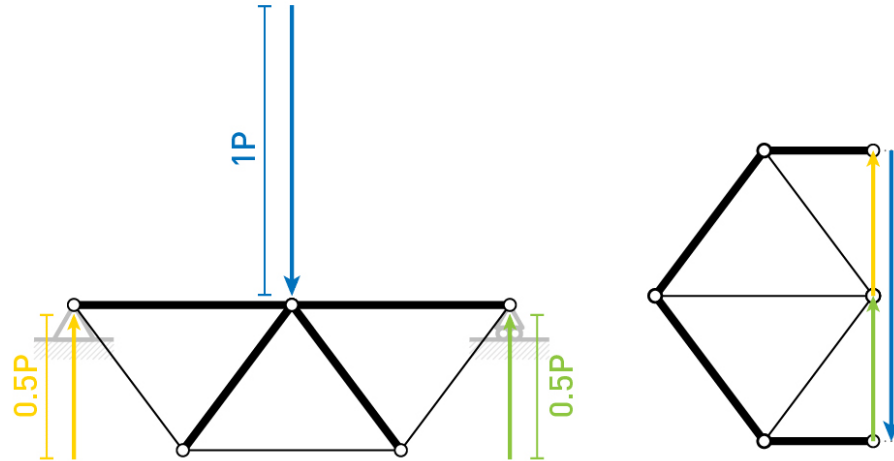
Generative design

- Subdivision graphic statics
- Shape grammars

Visualisation

...

$$\text{Minimum Volume} = \text{Minimum Load Path} \longrightarrow \min V = \min \frac{1}{\sigma} \sum |F_i| \cdot L_i$$



$$\begin{aligned} \Sigma FL &= 1.0 \\ \Sigma FL &= 1.0 \end{aligned}$$

Form analysis

Interactive (parametric)

- Analysis
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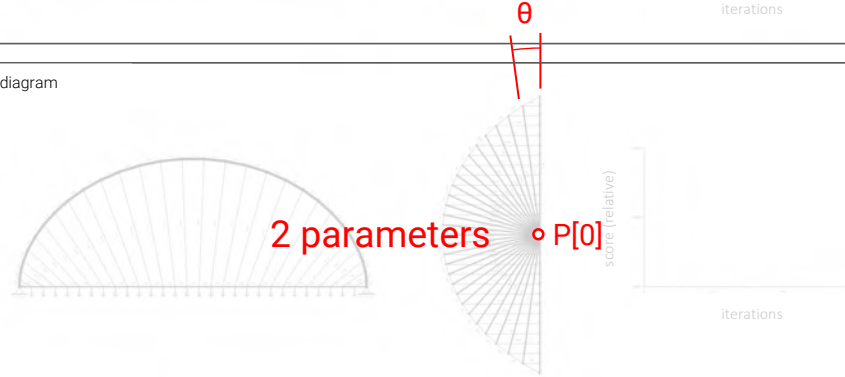
Optimization using numerical methods

14 parameters



Optimization using force diagram

2 parameters



Form analysis

Interactive (parametric)

- Analysis
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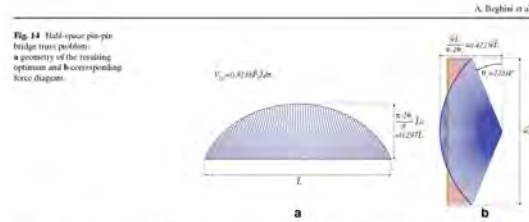
3D graphic statics

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point (see Appendix for a proof). Using the approximation in (17) we obtain:

$$\bar{H} = \frac{L}{4} \tan\left(\frac{\pi}{4} + \frac{\theta_1}{2}\right) \quad (36)$$

Therefore, the total volume of the truss in (37) using (17) is still described by only one parameter: the initial angle, θ_1 . The volume of the truss has been solved numerically and the final geometry and force diagrams are indicated in Fig. 14. The optimum is found for $\theta_1 = 22.04^\circ$, which corresponds to $\bar{H} = 0.371\bar{\omega}L$, $\phi = 36.58^\circ$ and a volume $V_{opt} = 0.8138PL/\sigma$. The depth of the truss and its dual based on (24) and (26) are: $\bar{h}_m = 0.2966L$ and $\gamma_m = 0.4215\bar{\omega}L$. The force diagram (Fig. 14b) shows that the force in the bridge horizontal tie is zero at the quarter points, consistent with the results presented by McCormack (1974).

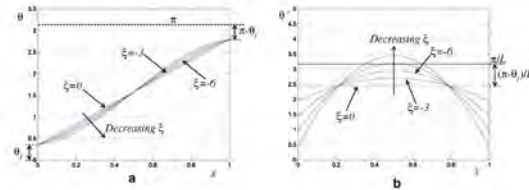


Fig. 15 a Plot of the expanded approximation for function $\theta(x)$ and b its derivative

Springer

Layout of a least weight single span structure with uniform load

Table 3 Optimal bridge truss with finite number of bars (small red circles indicate the material addition from Beghini et al. 2013a)

Number of Bars	Form Diagram using (17)	Force Diagram using (17)	$\sigma V_{opt}/PL$ from Beghini et al. (2013a)	$\sigma V_{opt}/PL$ using (17)	$\sigma V_{opt}/PL$ using (17)
1			1.175 ($\theta_1 = 4.34^\circ$)	1.175 ($\theta_1 = 4.34^\circ$)	1.175 ($\theta_1 = 4.34^\circ$)
8			1.119	1.119 ($\theta_1 = 0.9^\circ$)	1.119 ($\theta_1 = 0.9^\circ$)
12			1.071	1.071 ($\theta_1 = 0.0^\circ$)	1.071 ($\theta_1 = 0.0^\circ$)
16			1.049	1.049 ($\theta_1 = -0.002^\circ$)	1.049 ($\theta_1 = -0.002^\circ$)
24			1.029	1.029 ($\theta_1 = -1.3^\circ$)	1.029 ($\theta_1 = -1.3^\circ$)

Springer

Form analysis

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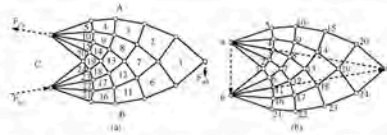
Subdivision graphic statics

Shape grammars

Visualisation

Maxwell's reciprocal diagrams and discrete Michell trusses

Fig. 5 The solution to the discrete Michell cantilever (Chan 1966; Maarek et al. 2011) in self-reciprocal form diagrams. Force diagrams. The geometry of the form and force diagrams are the same



diagrams in Fig. 2 is also the geometry of a second discrete optimal truss with its own loading is explored in Fig. 3. The geometry of the truss in Fig. 3 (the form diagram) is taken from the force diagram in Fig. 2.

Now that two optimal truss geometries have been found, the remaining task is to find the external loads that correspond to a truss represented by the geometry of the force diagram. The external forces for this second truss are found by closing the "open" polygons of the original form diagram in a continuous manner. The vector formed by closing an open polygon in the original form diagram is the external force that would be applied to the node in the force diagram that corresponds to the original open polygon. Closing the open polygons provides the proper external forces for the dual structure because it closes the force polygon at the node which corresponds to the open polygon in the original form diagram.

It is clear that the form and force reciprocal diagrams in Figs. 2 and 3 are interchangeable. In order to make the relationships more clear, Ross's notation in Fig. 5 is reversed.

For the problem in (I), the lengths are taken from the truss on the left and the internal forces from the truss on the right, though the reverse (i.e. internal forces from the left diagram, lengths from the right diagram) would result in the same optimal solution. The advantage to using Graphic Statics for this class of optimal problems is that it provides all of the information about the loads and the results in a graphical manner.

Another example of paired or dual discrete trusses is given in Fig. 4, where the discrete truss on the left (form diagram) is the minimum load path structure for a given set of

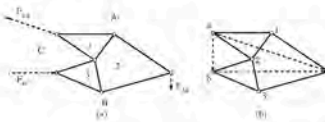
forces and nodal connectivity (Prager 1970). The reciprocal force diagram in the middle not only represents the forces of truss but also the geometry of another optimal truss, where the external forces for this optimal structure can also be found by closing the open polygons in the form diagram. For example, the external force applied at node a in Fig. 4b is given by closing the "open" polygon, A , or drawing a line from joint A -3-C-A to joint A -B-D-A. Figure 4c shows the dual truss with the external loads given by closing all of the polygons in (a).

The Appendix has a chart showing some dual discrete trusses along with the flow notation. This chart graphically shows both the geometry and the forces for each pair of optimal trusses.

4 Self-reciprocal discrete Michell trusses

For some cases of paired trusses, it can be shown that the geometry of the reciprocal force diagram is the exact same as the member diagram. It can be said that these types of structures are self-reciprocal. An example of a self-reciprocal structure is the optimal discrete Michell truss taken from Chan (1969) and Maarek et al. (2011) shown in Fig. 5. This self-reciprocity can be observed for an unbounded cantilever with an equal number of members at each support when there is a single applied load acting parallel to a line drawn through the supports and the reaction member the top of the cantilever. Figure 6 shows another example of a self-reciprocal frame for the three-point problem. We note here that the notion of self-reciprocity can also apply for non-symmetric structures. The reasons for

Fig. 6 The solution to the three-point discrete truss problem (off the optimal or form diagram) is force diagrams



Springer

Maxwell's reciprocal diagrams and discrete Michell trusses

Appendix

The following table presents a summary of the reciprocal relationships between the form and force diagrams of several optimal structures.

	Form diagram	Force diagram	Dual truss
I.			
II.			
III.			
IV.			

I. Classical discrete Michell solution for the centrally-loaded beam (Michell 1904; Hemp 1973; Maarek et al. 2011)

II. Two discrete Michell solutions for semi-infinite space (Michell 1906; Hemp 1973)

III. Self-reciprocal cantilever with two points of support (Chan 1966; Maarek et al. 2011)

IV. Optimal shear bracing solution (Hemp 1973; Sorensen et al. 2012). Note: Overlapping lines have been merged

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Interactive (parametric)

Analysis

Constraint-driven design

Computational graphic statics

Algebraic graph(ic) statics (AGS)

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Optimization

Form generation

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Thrust network analysis (TNA)

3D graphic statics

Generative design

Subdivision graphic statics

Shape grammars

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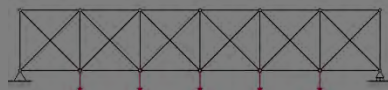
Table 2 Optimization of the roof truss

Description	Form Diagram	Force Diagram	Normalized Volume
(a) Initial truss connectivity			1.000
(b) Benchmark truss: top chord, cantilevers and web members unconstrained			0.552
(c) Chord profiles constrained for architectural and functional reasons			0.629
(d) Depth constrained, straight web members (X-diagonals)			0.852
(e) Depth constrained			0.669

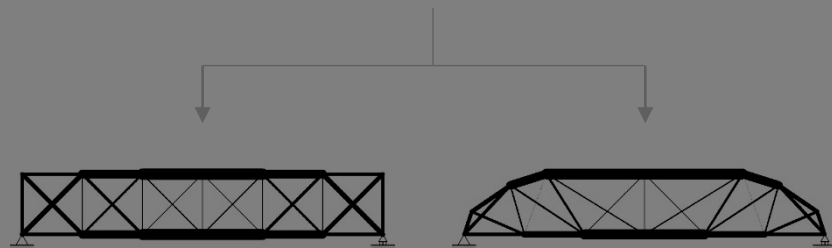
loads or the supports can change as the optimal solution is determined.

The authors are currently exploring other applications where the design variables are in the force domain.

Form analysis



Pre-determined Shape



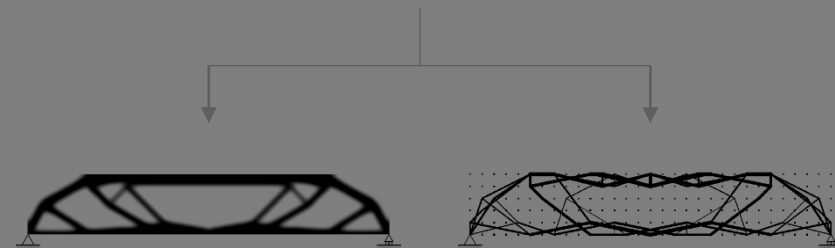
Size

Shape / Geometry

Form generation



Unknown Shape



Topology (Continuum)

Topology (Discrete)

Form analysis

Interactive (parametric)

- Analysis
- Constraint-driven design

Computational graphic statics

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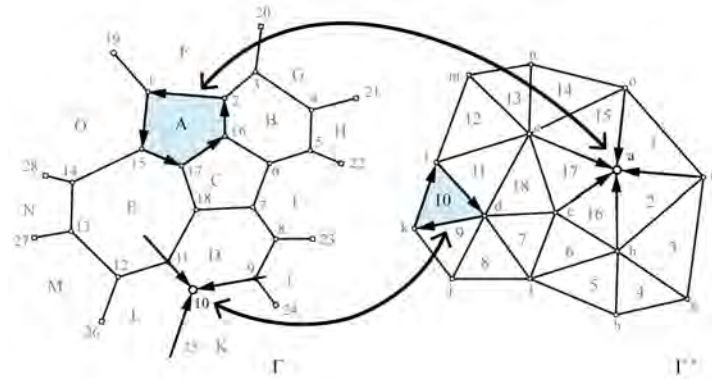
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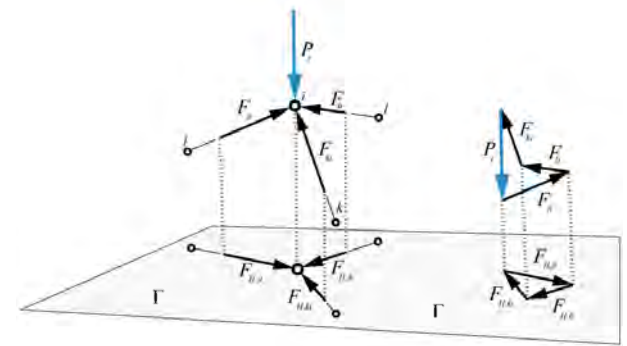
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Form Diagram

Force Diagram



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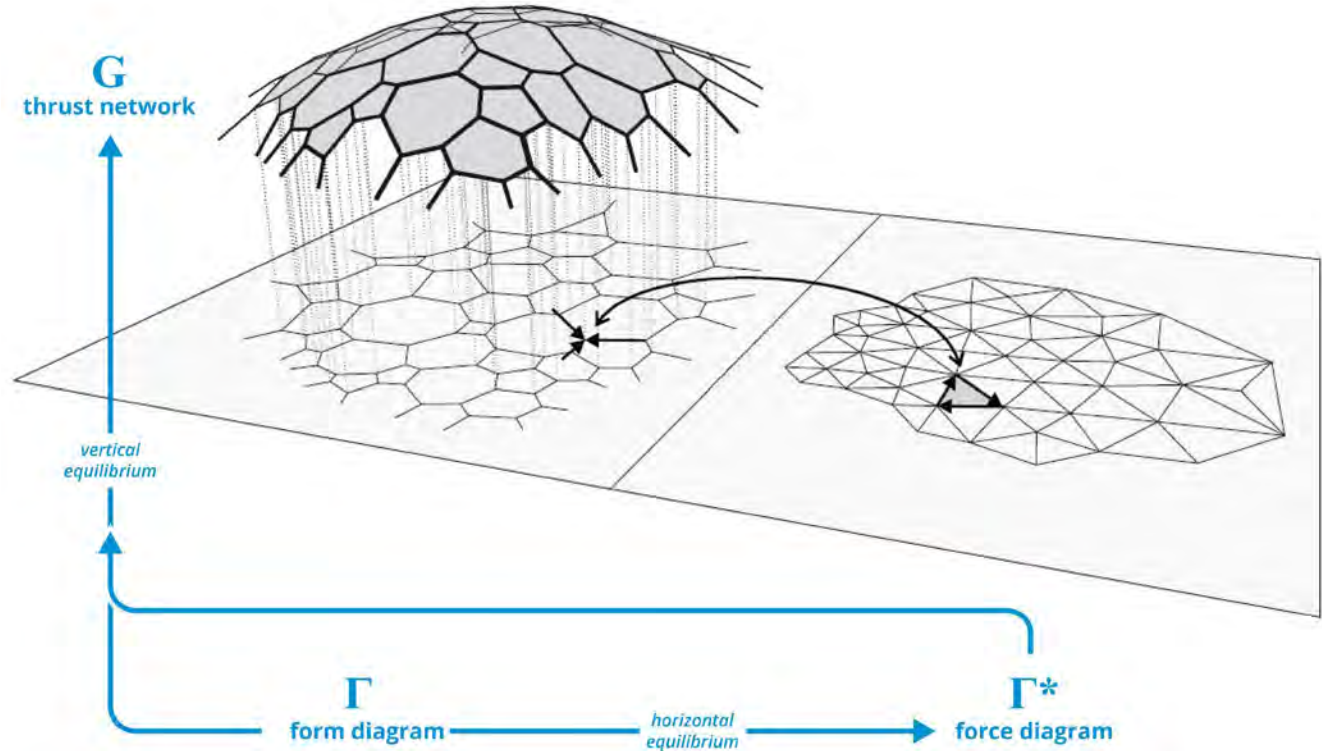
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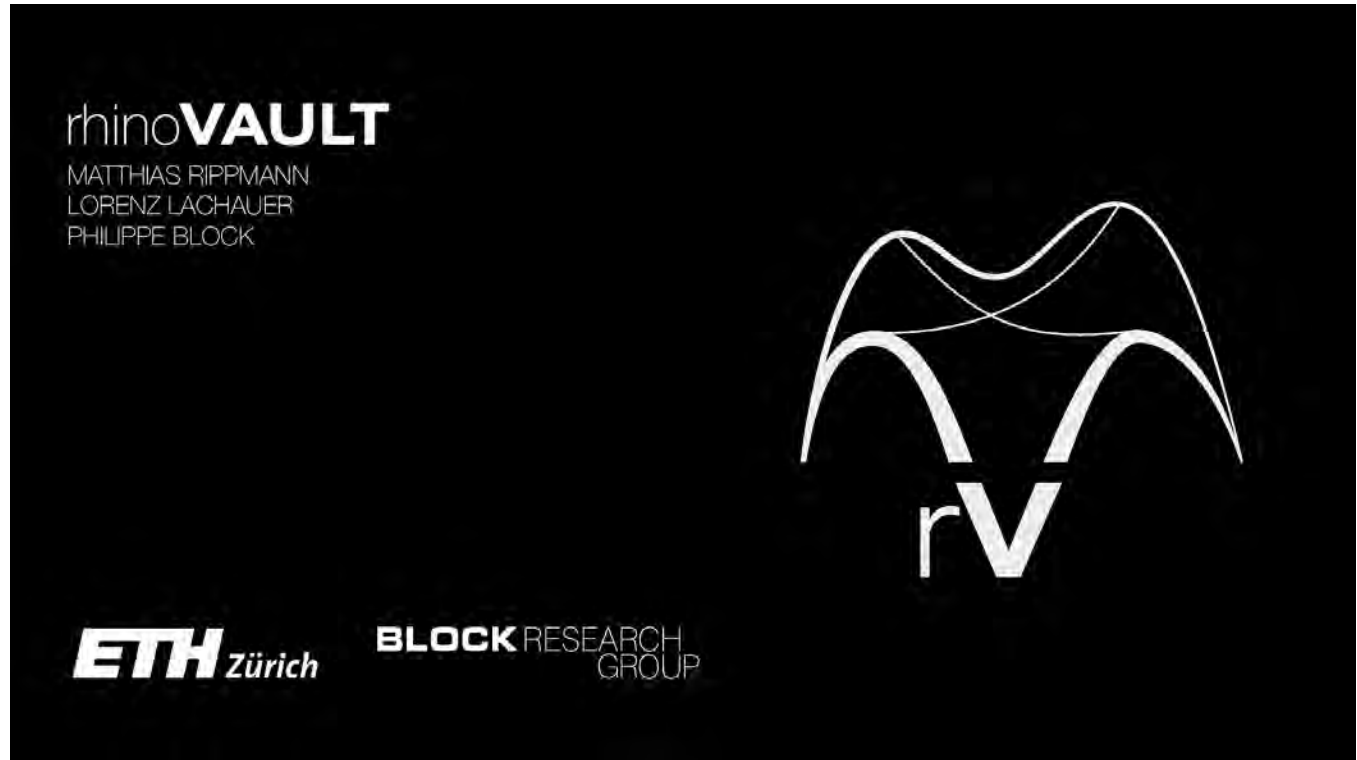
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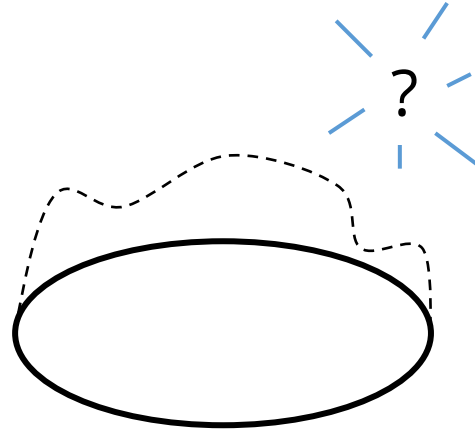
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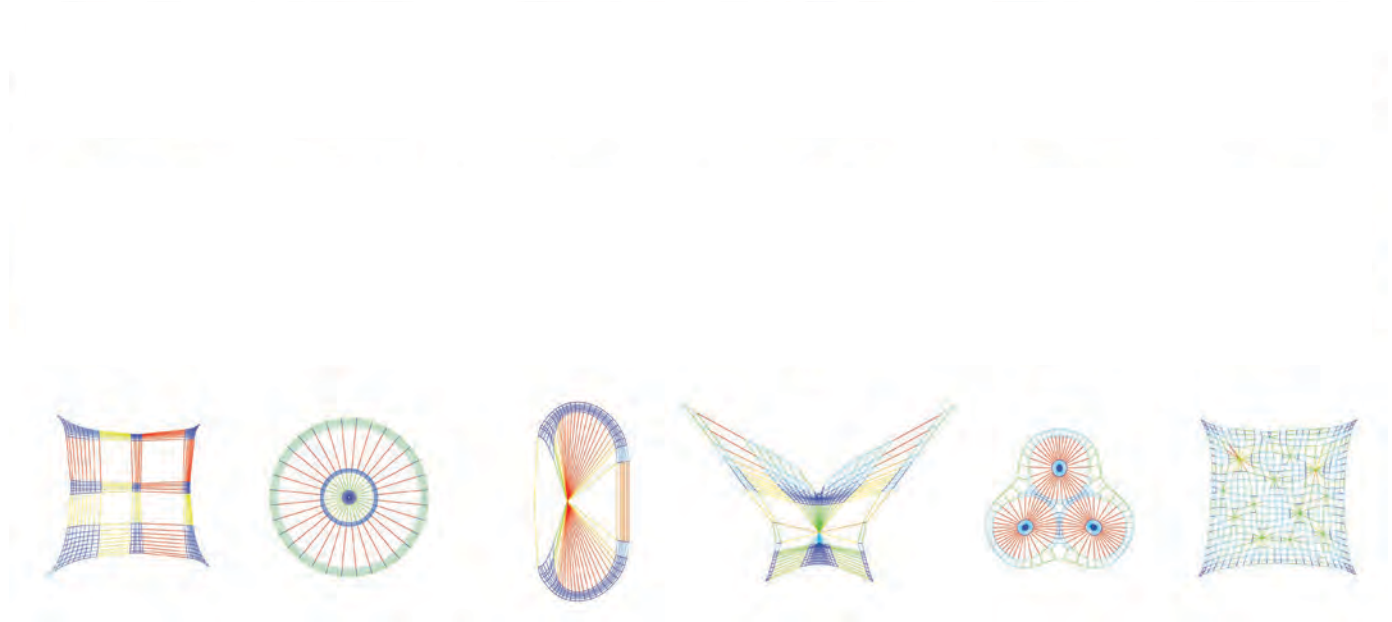
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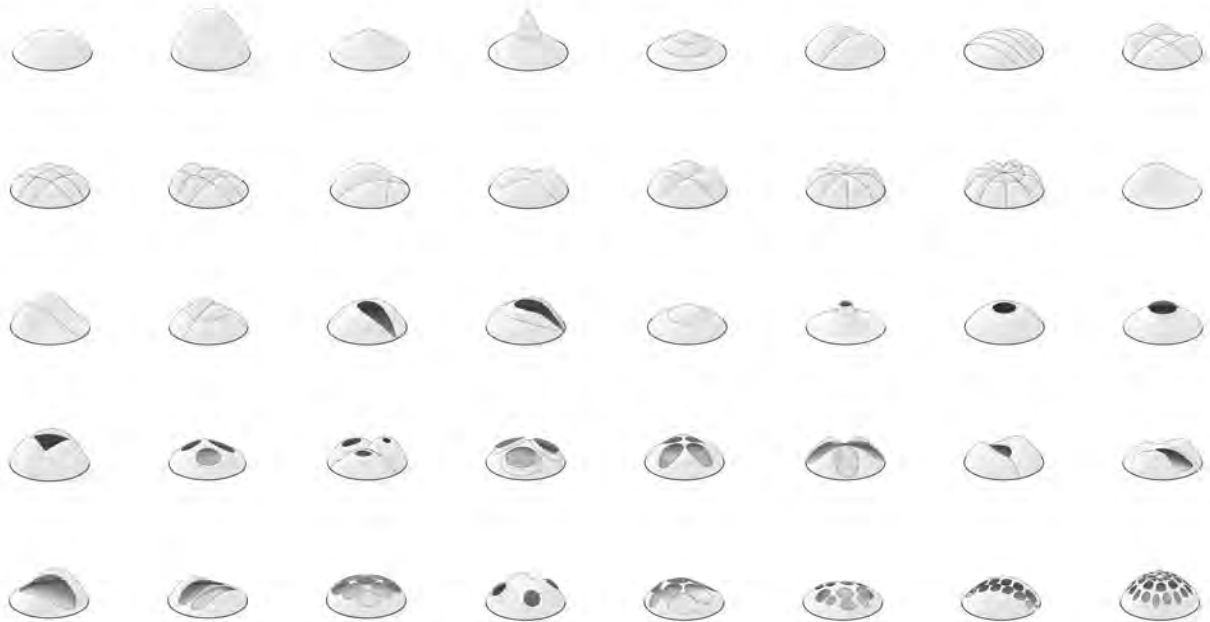
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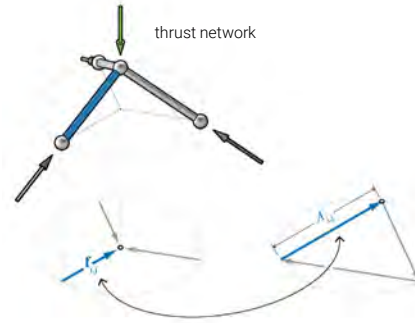
...



form diagram

force diagram

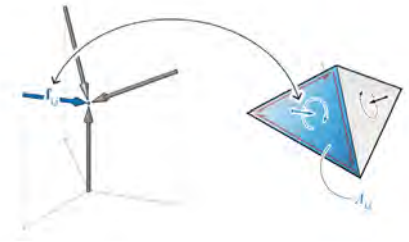
2D



form diagram

force diagram

2.5D



form diagram

force diagram

3D

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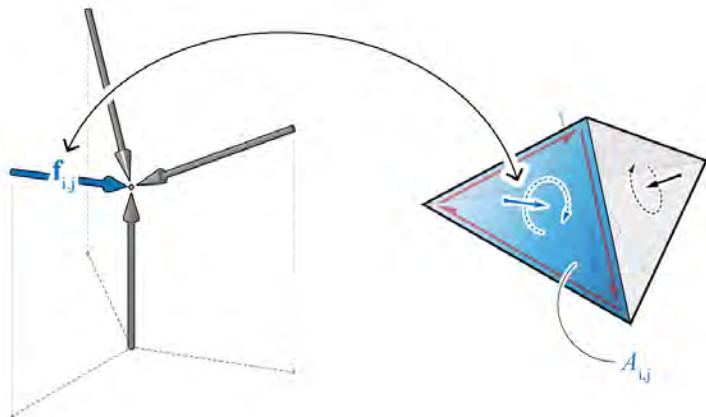
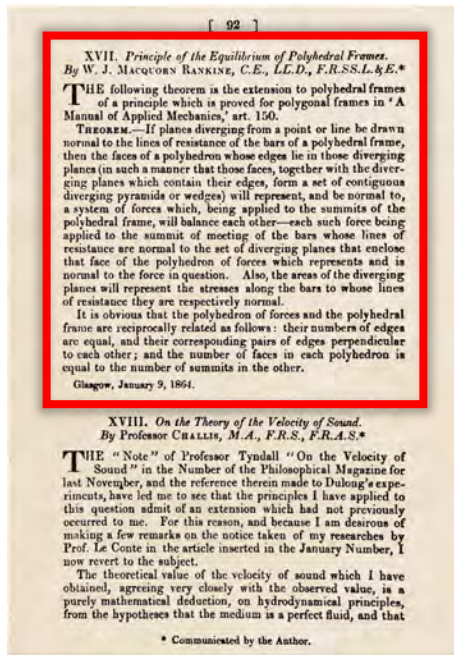
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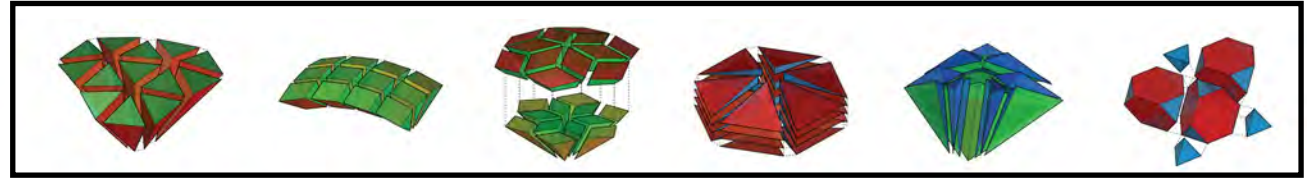
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Polyhedral force diagram



Polyhedral form diagram

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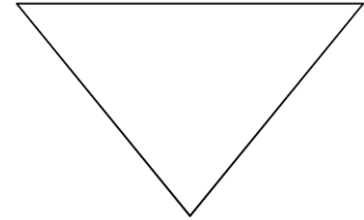
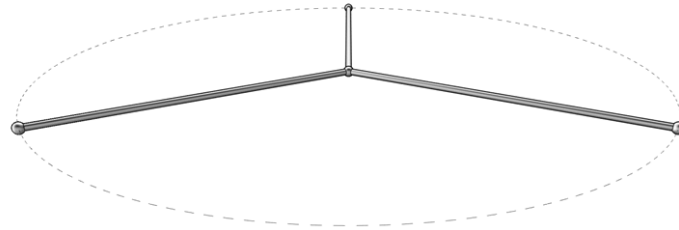
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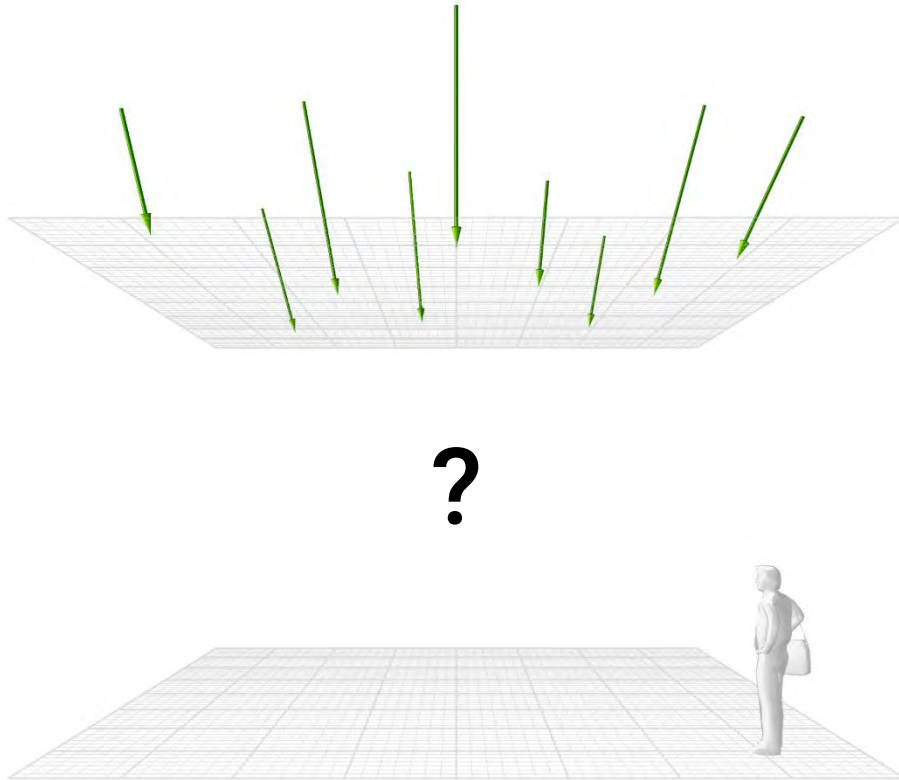
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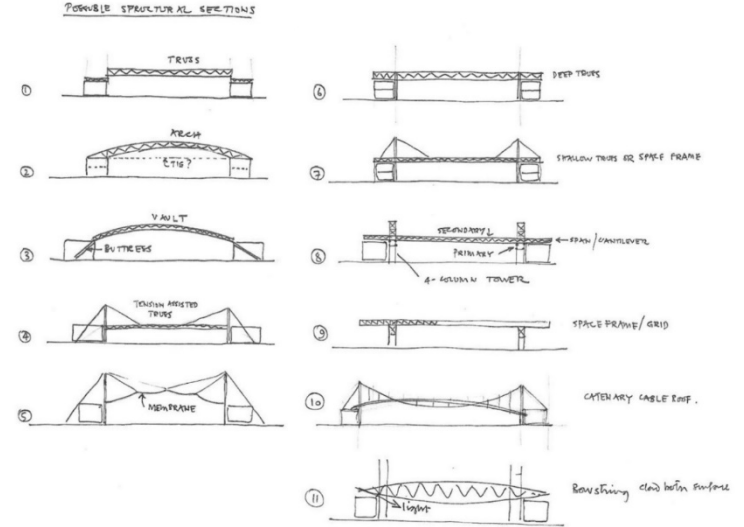
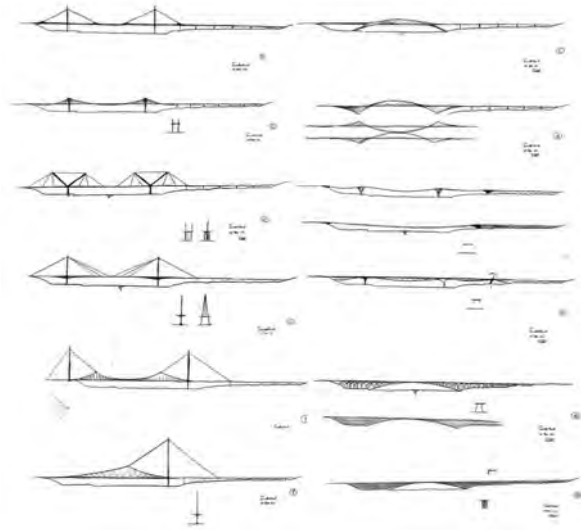
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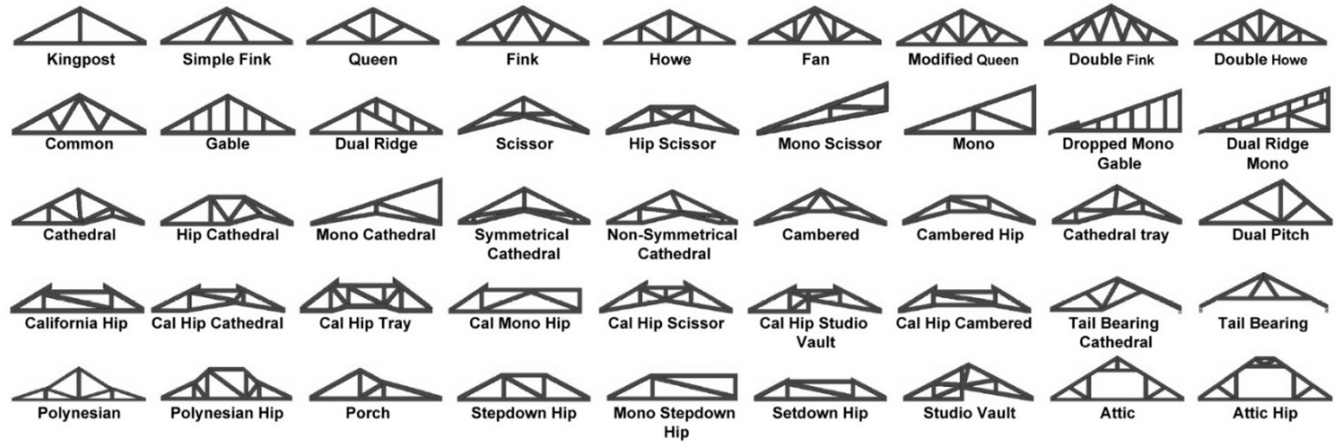
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“In order to arrive at new forms and concepts,
we have to free ourselves from such pre-defined typologies.”

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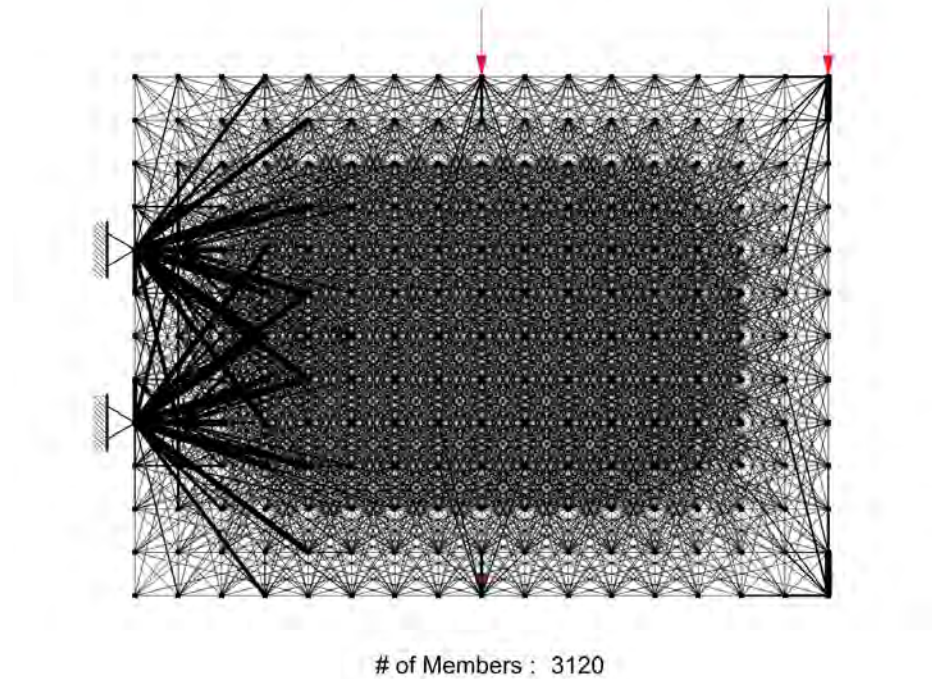
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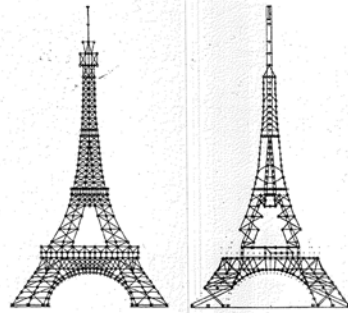


Fig. 4.5. The flexibility in choice of ground structure. Optimal design of a well-known structure. Left hand picture shows the ground structure and the right hand picture the optimal topology for a single downward load at the top of the structure. The example shows that it is reasonable to consider multiple load cases for realistic structures. By courtesy of M. Kocvara and J. Zowse.

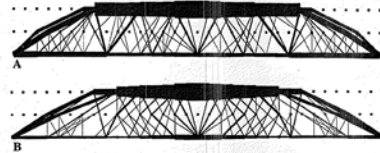


Fig. 4.6. The difference (and similarity) between multiple load case treated in the weighted average formulation (equal weights) (A) and treated in the worst case min-max formulation (B). Optimal truss topologies for transmitting three vertical forces to two fixed supports as in Fig. 4.7A (cf. Fig. 4.2), but for a long slender rectangular ground structure of aspect ratio 16 (like a long span bridge), with 33 by 3 equidistant nodes and all 2818 possible non-overlapping connections. In the figures, the vertical scale has been distorted in order to be able to show the results. Bendsoe, Ben-Tal and Zowe, 1993.

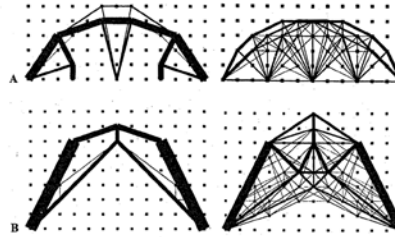


Fig. 4.7. The difference between multiple load and single load case problems. Optimal truss topologies for transmitting three vertical forces to two fixed supports. Two positions of the loads are considered and the trusses are optimized with the loads treated as a single load as well as three individual load cases for a min-max, worst case design situation. The ground structures consist of all 8744 possible non-overlapping connections between the nodal points of a regular 13 by 13 mesh in a square domain. The loads are vertical unit loads at three equidistant nodes along the lower line of nodes (A) or across the middle of the ground structure (B) (see also Fig. 4.2). The left hand column shows the single load results, the left hand column the multiple load, worst case results. We do not show the uppermost rows of nodes, as these are not part of the optimal structure. A slight asymmetry of the ground structure is reflected in the optimal truss topologies. Bendsoe, Ben-Tal and Zowe, 1993.



Fig. 4.8. A detailed study situation of Fig. 4.7A, but with only the mid-span load applied. The ground structure is restricted to a rectangular domain of aspect ratio 2, and with the number of nodes increased to a 21 by 11 layout, with 16290 possible non-overlapping connections. Note that the supports have been moved in by two nodes from each vertical side, in order to identify an eventual restriction of having the supports at the extreme points of the ground-structure. Bendsoe, Ben-Tal and Zowe, 1993.

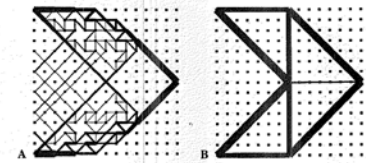


Fig. 4.9. A case where the introduction of multiple loads simplify the optimal lay-out. Also an example of the optimal topology for a ground structure with only neighbouring nodes in a square, regular 15 by 15 layout being connected (see Fig. 4.1B); this results in only 788 potential bars. All nodes at the left hand side are potential supports (see Fig. 4.2). (A) The optimal topology for a single vertical load at the mid right hand node. (B) The optimal topology for three load cases including the load of the single load problem. The load of the single load example is twice as large as the two other loads, one of which is a horizontal load at the mid right hand node while the last load is a vertical load at the centre of the ground structure. This is for a weighted average formulation with equal weights. The compliance for the load case number 1 increases by only 2.6%, as compared to (A), which is optimal for this load only. Bendsoe, Ben-Tal and Zowe, 1993.

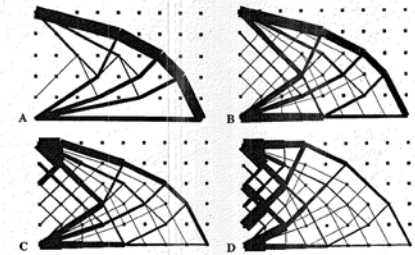


Fig. 4.10. The effect of self-weight loads. Optimal truss topologies for transmitting a single vertical force to a vertical line of supports (see Fig. 4.2). The figures show the variation for increasing specific self-weight loads, corresponding to increasing real lengths of the structures. In (A) self-weight is ignored, in (B) self-weight is present, increased by 2 times to the design (C). These designs are obtained for a 9 by 6 equidistant nodal layout in a rectangular domain of aspect ratio 1.6, and all 919 possible non-overlapping connections. If all 1431 possible connections are used, the design (C) is modified to the design (D). Bendsoe, Ben-Tal and Zowe, 1993.

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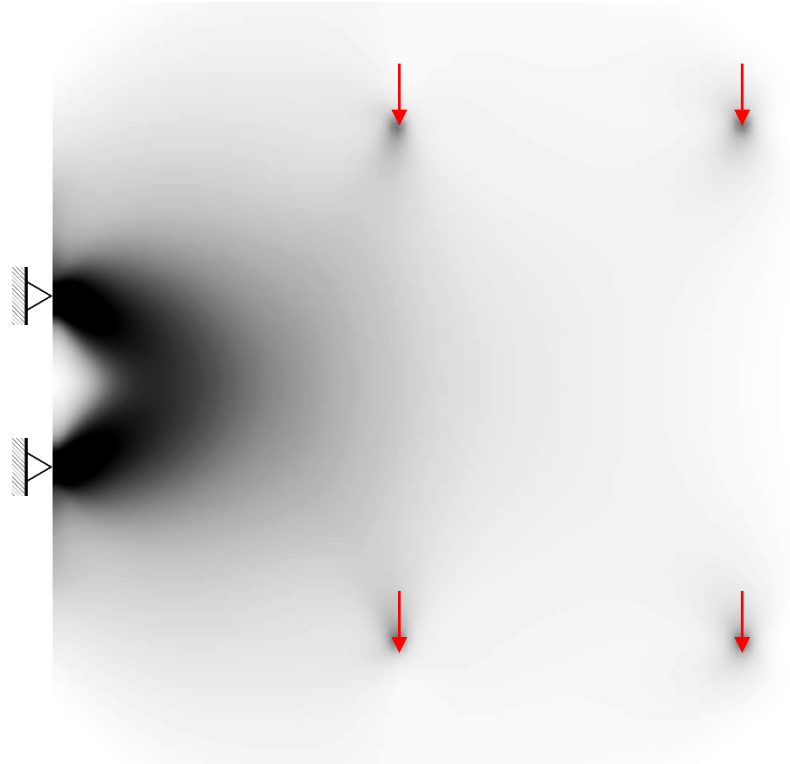
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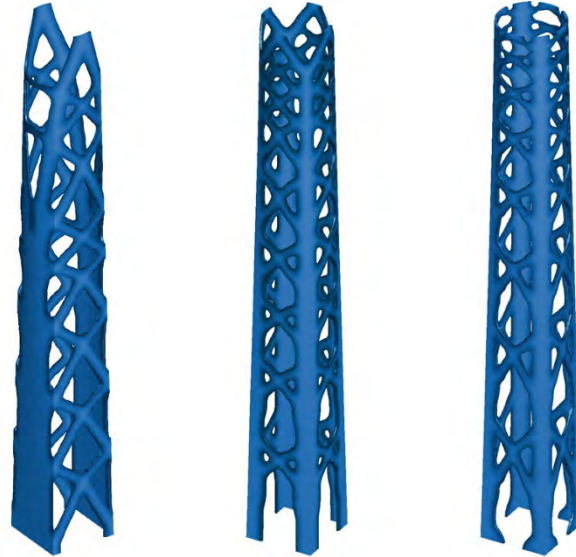
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SCALE

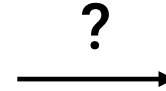
Unrealistic member thickness / stiffness !

CONTINUITY

Fabrication & constructability issues !

PIXELATION

Unclear location of intersections (joints)



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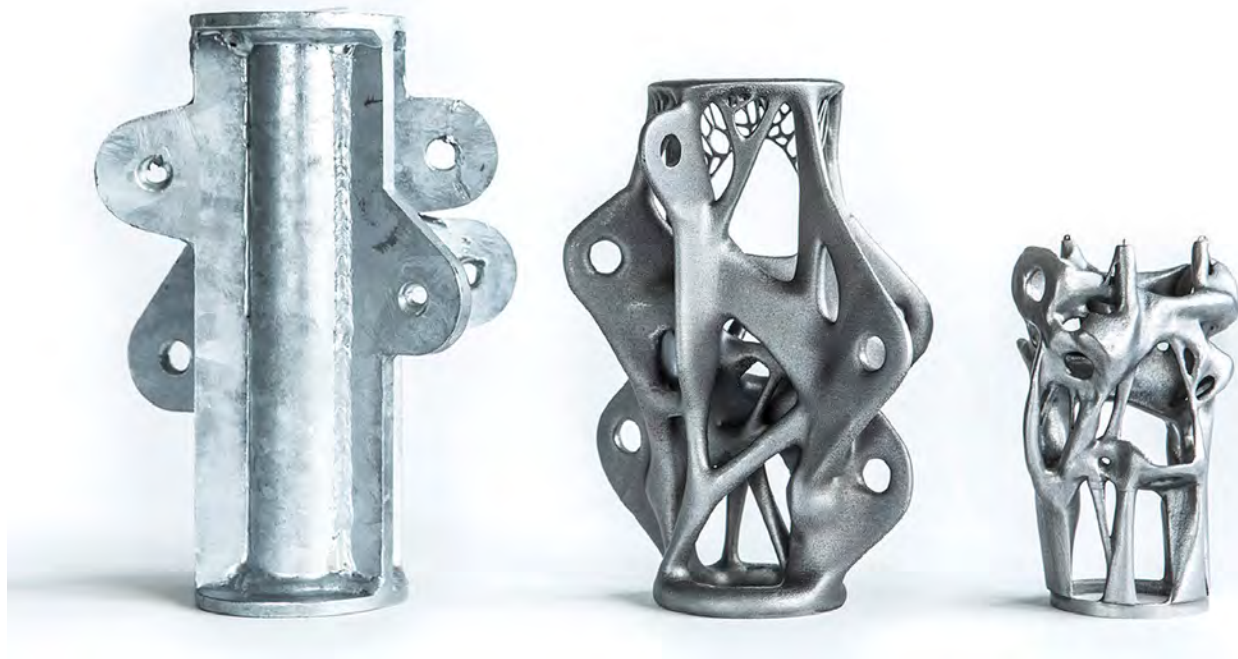
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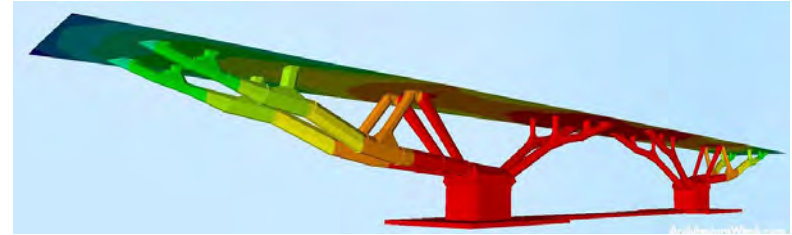
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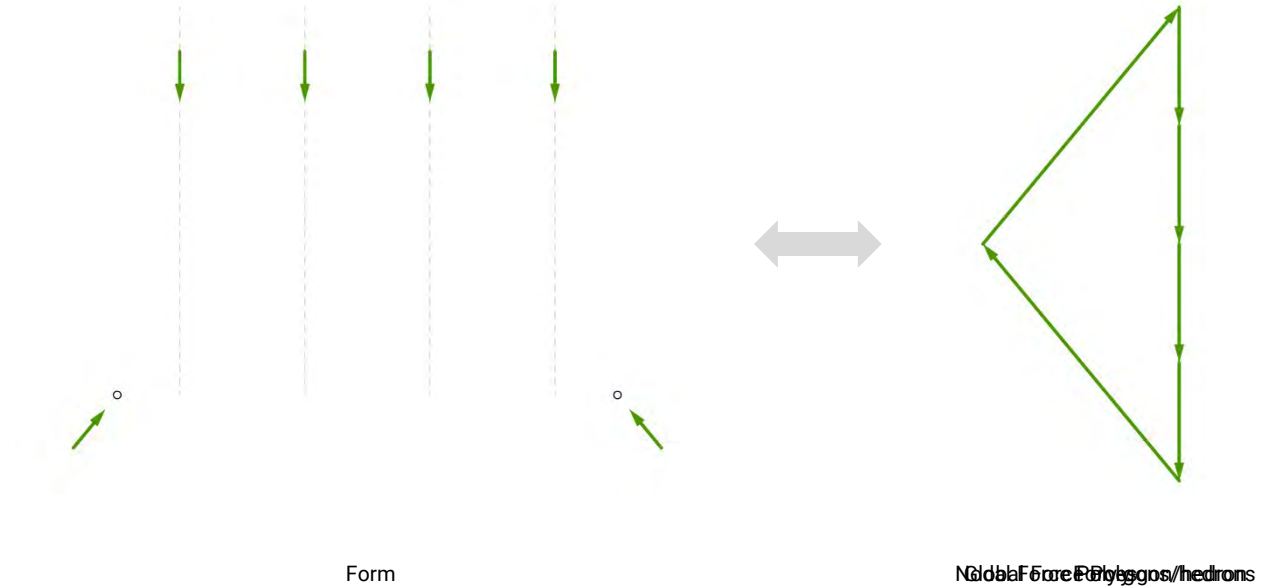
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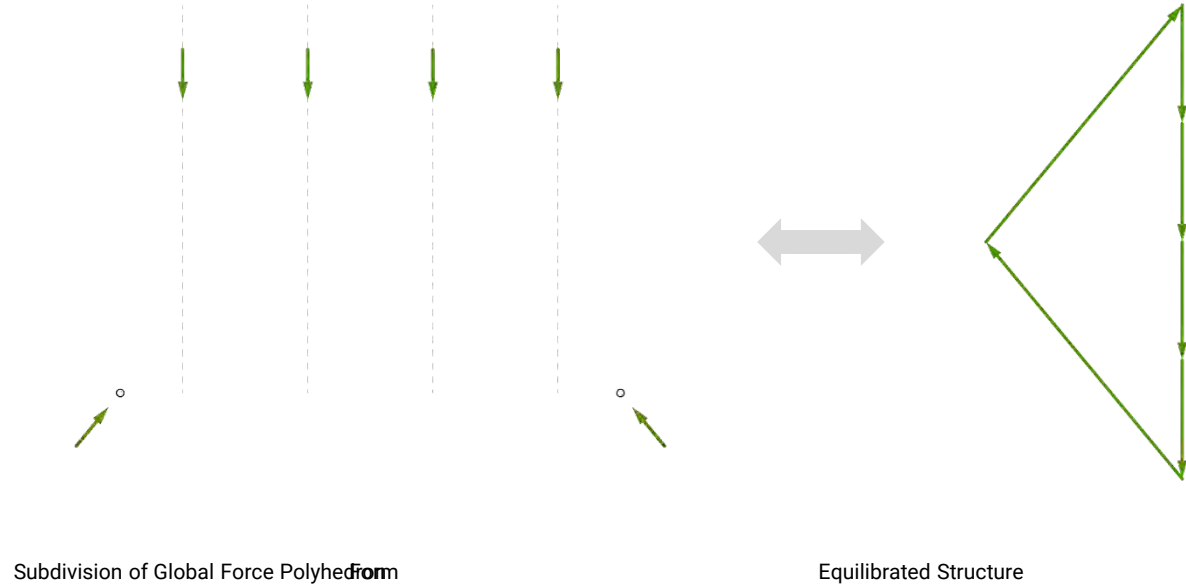
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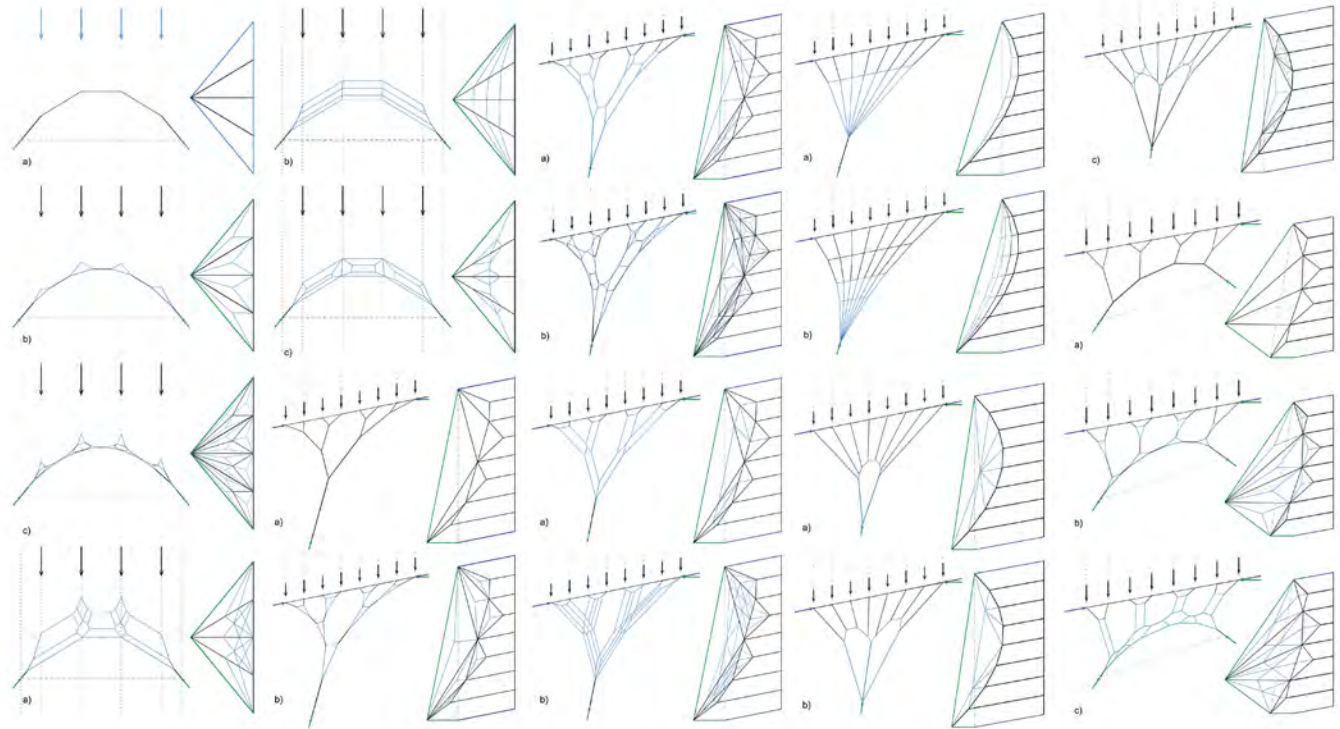
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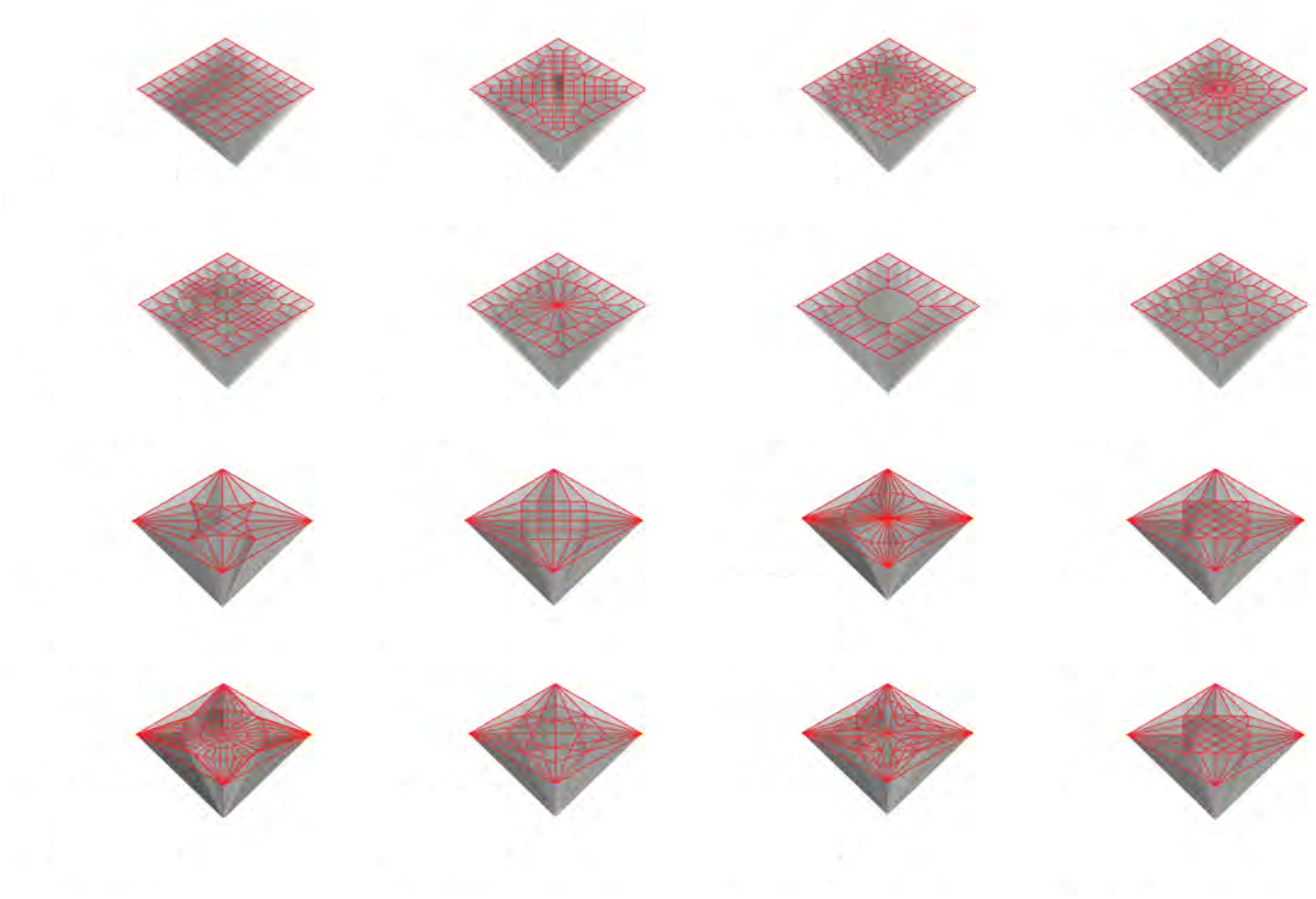
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Thrust network analysis (TNA)
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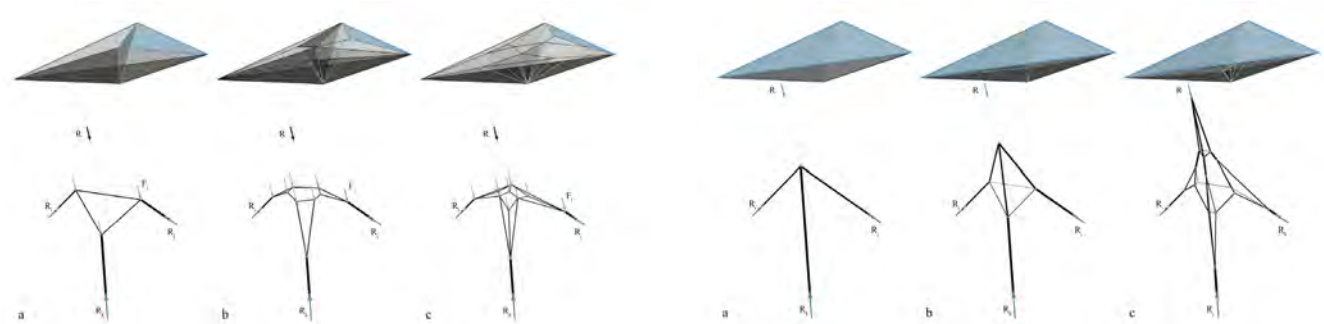
Generative design

[Subdivision graphic statics](#)

Shape grammars

Visualisation

...



Subdivision of a Face

Subdivision of Cells

Form analysis

Interactive (parametric)

- Analysis
- Constraint-driven design

Computational graphic statics

- Algebraic graph(ic) statics (AGS)
- Bi-directional AGS

Optimization

Form generation

Form finding

- Thrust network analysis (TNA)
- 3D graphic statics

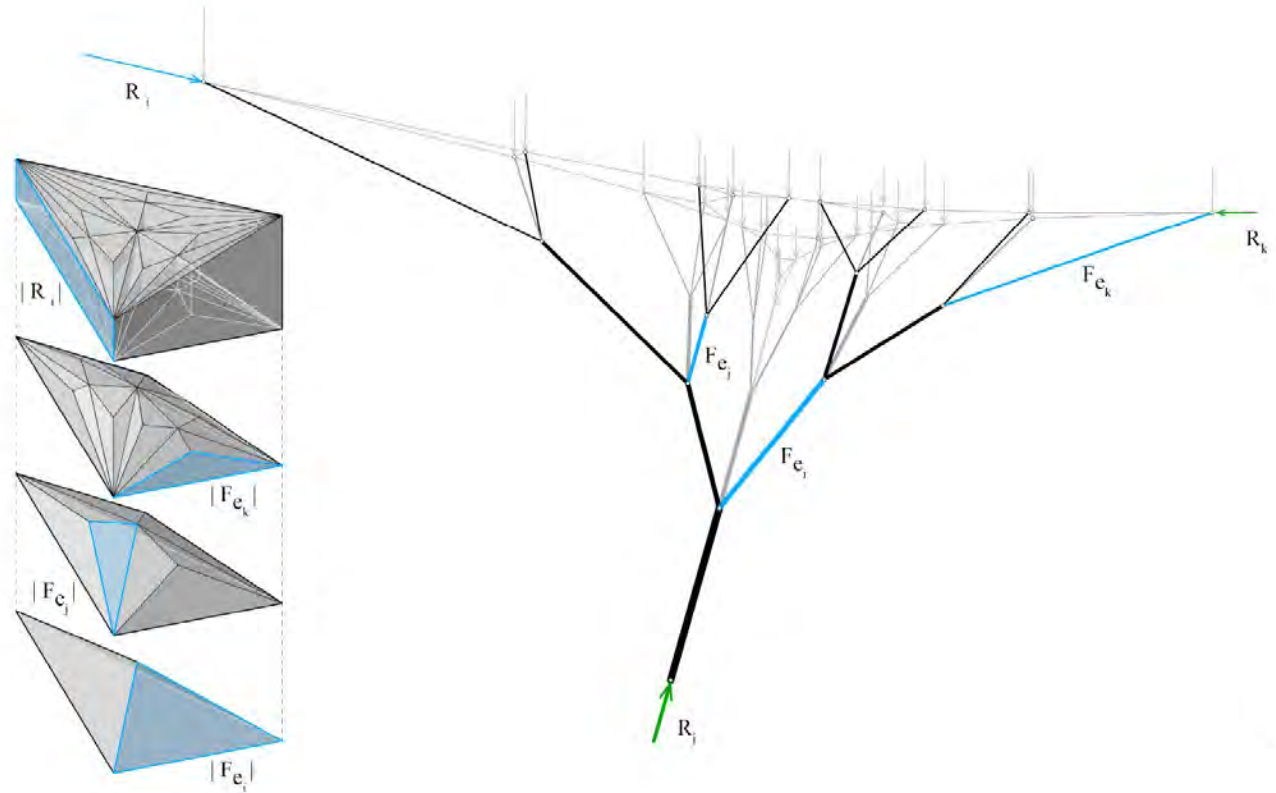
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Subdivision graphic statics

- Shape grammars

Visualisation

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Form analysis

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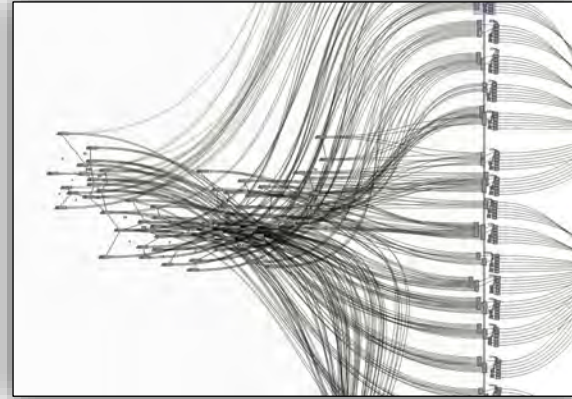
- Thrust network analysis (TNA)
- 3D graphic statics

Generative design

- Subdivision graphic statics
- [Shape grammars](#)

Visualisation

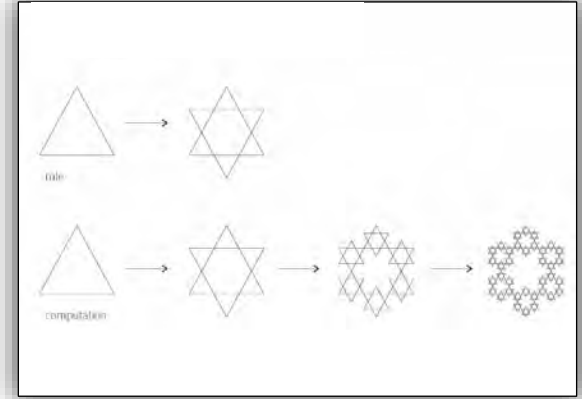
...



Parametric Design Space

NUMERICAL PARAMETERS are defined ...
to
... **DESCRIBE & MODIFY** geometries.

VS



Grammatical Design Space

ACTIONS & OPERATIONS (RULES) are defined ...
to
... **GENERATE** geometries.

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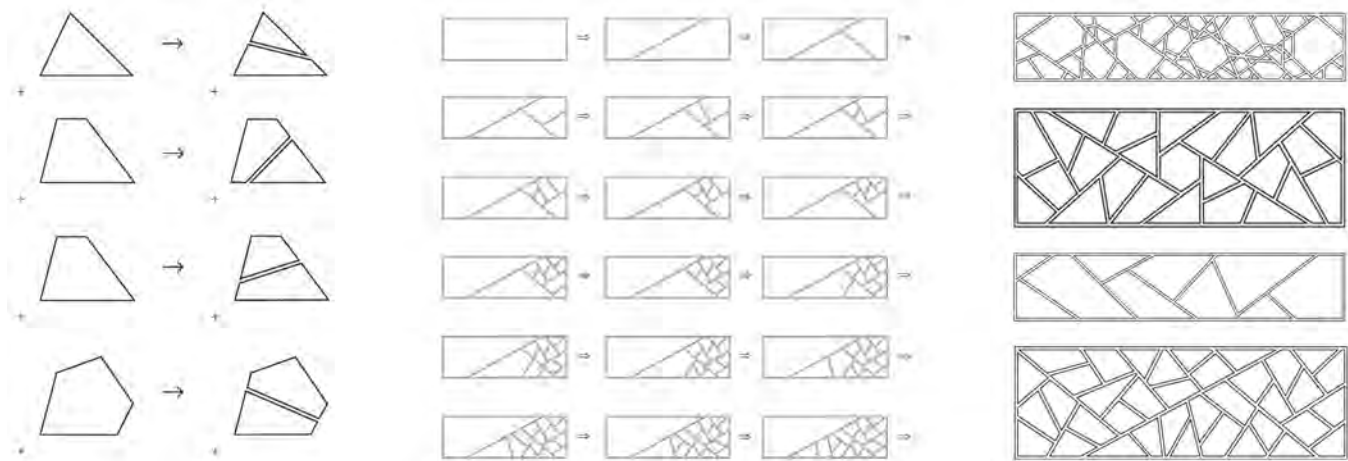
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[Shape grammars](#)



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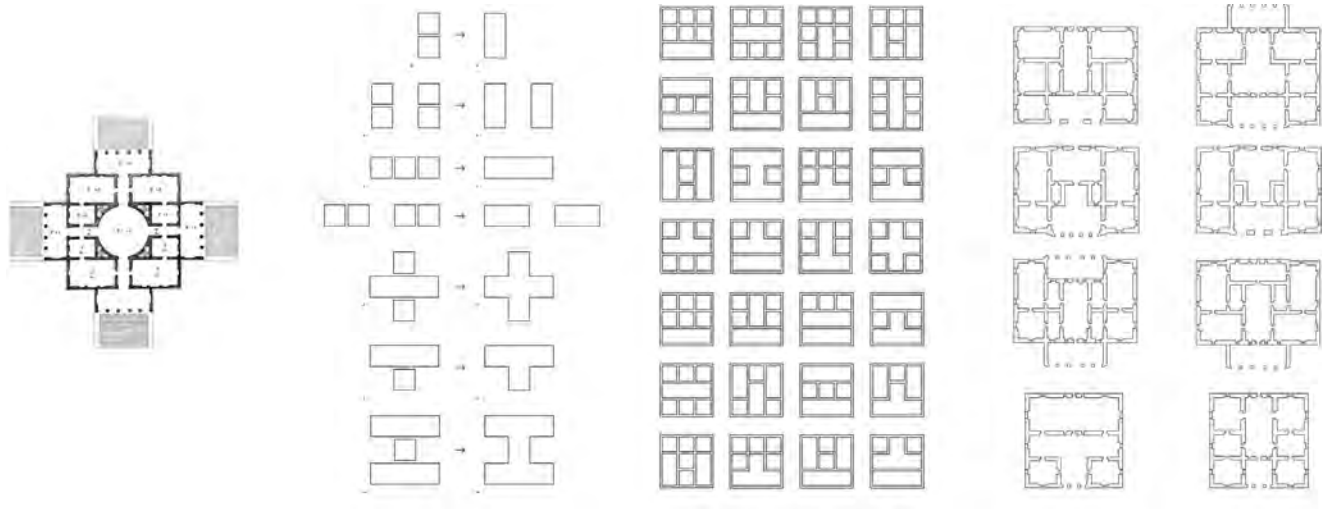
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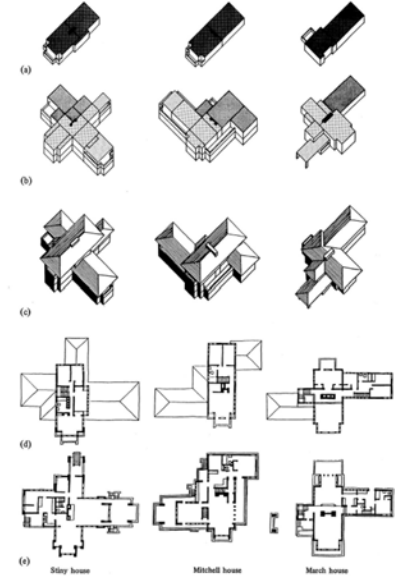
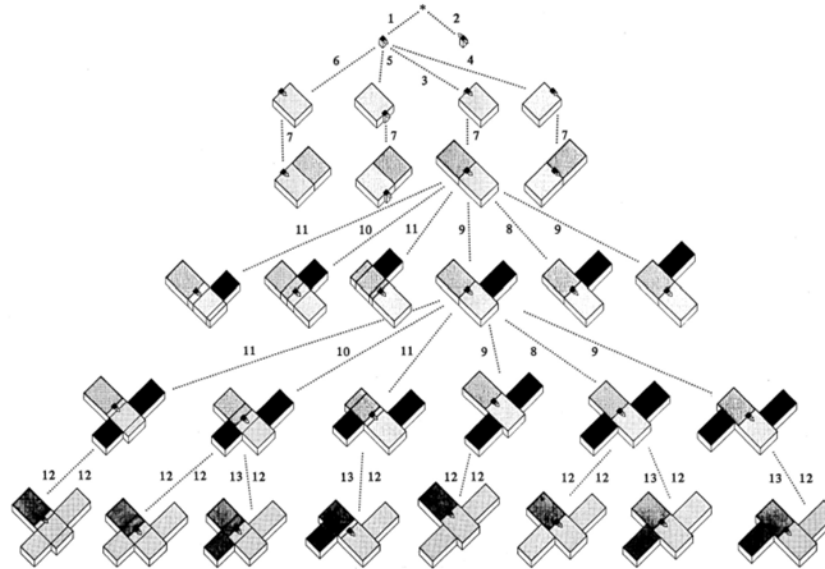
Thrust network analysis (TNA)
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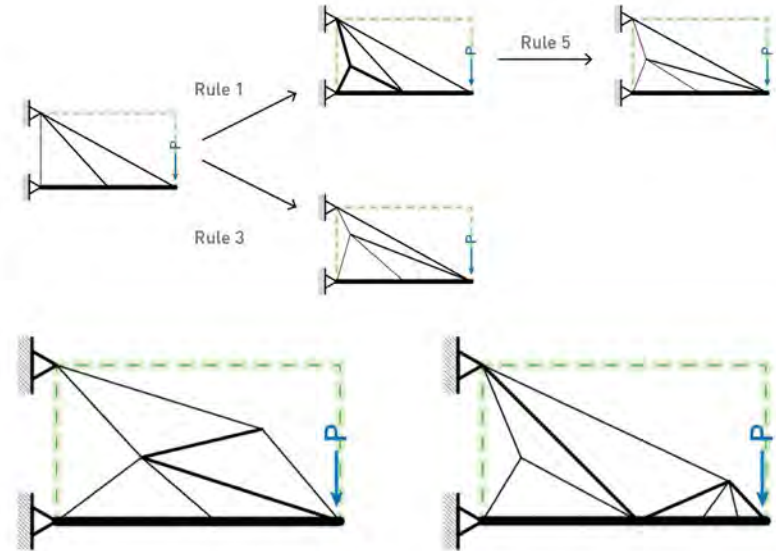
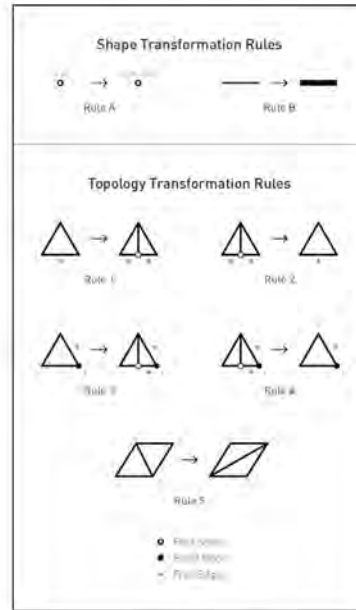
Thrust network analysis (TNA)
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Visualisation



Original Work by John S. Cagan (1999) • Revisited & 1

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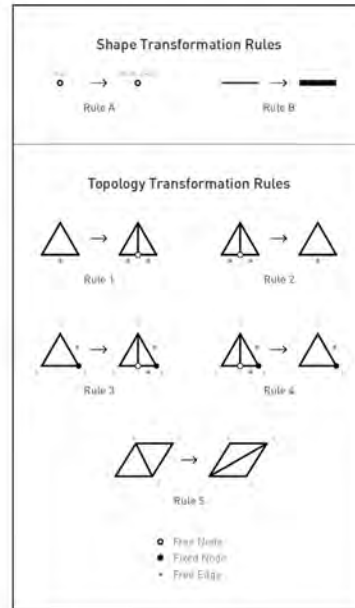
3D graphic statics

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Subdivision graphic statics

[Shape grammars](#)

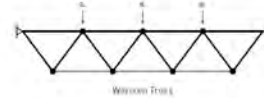
Visualisation



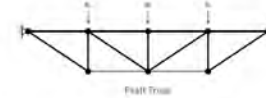
Design Problem



Conventional Solutions

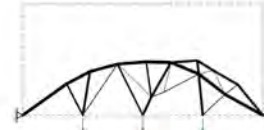


100%



97.0%

Shape Grammar Solutions



90.2%



89.4%



88.4%

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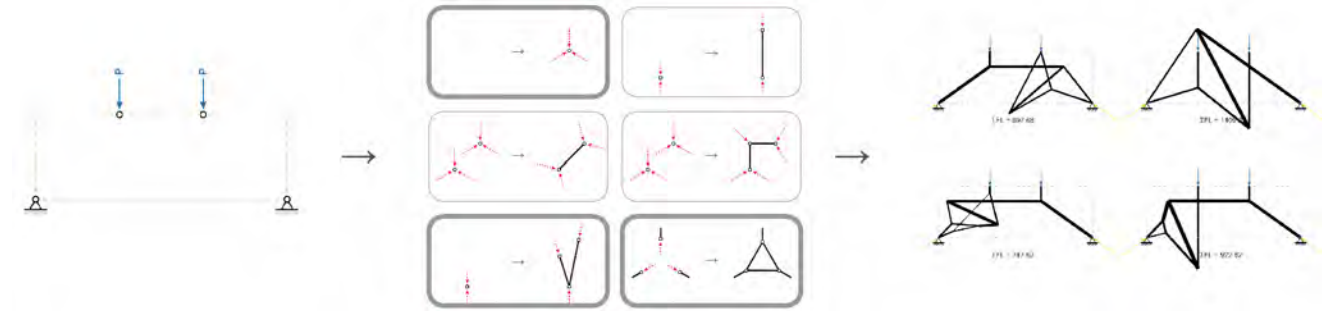
Form finding

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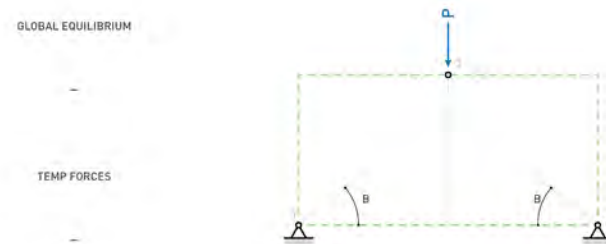
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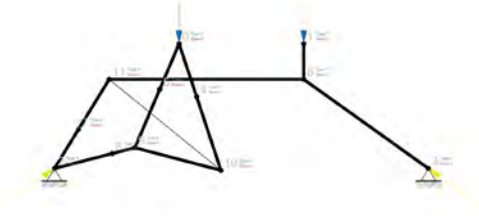
Visualisation

...

Force Diagrams of Every Node



The Design



Evaluation Metric

ID : 0004E
 $\Sigma FL = 673.61$
 $\Sigma L = 126.38$
 $\Sigma \# = 13$

Rule Info

Rule 1: 25%
Rule 2: 25%
Rule 3: 25%
Rule 4: 25%
Rule 5: 25%

RULE HISTORY

1. "Rule 0: START" applied at Node 2
2. "Rule 3: Split Node" applied at Node 1
3. "Rule 5: Connect via Projection" applied at Node 4
4. "Rule 5: Connect via Projection" applied at Node 1
5. "Rule 3: Split Node" applied at Node 3
6. "Rule 5: Connect via Projection" applied at Node 8
7. "Rule 6: System Close" applied at Node 4
8. "Rule 6: System Close" applied at Node 10
9. "Rule 6: System Close" applied at Node 6

Rule History

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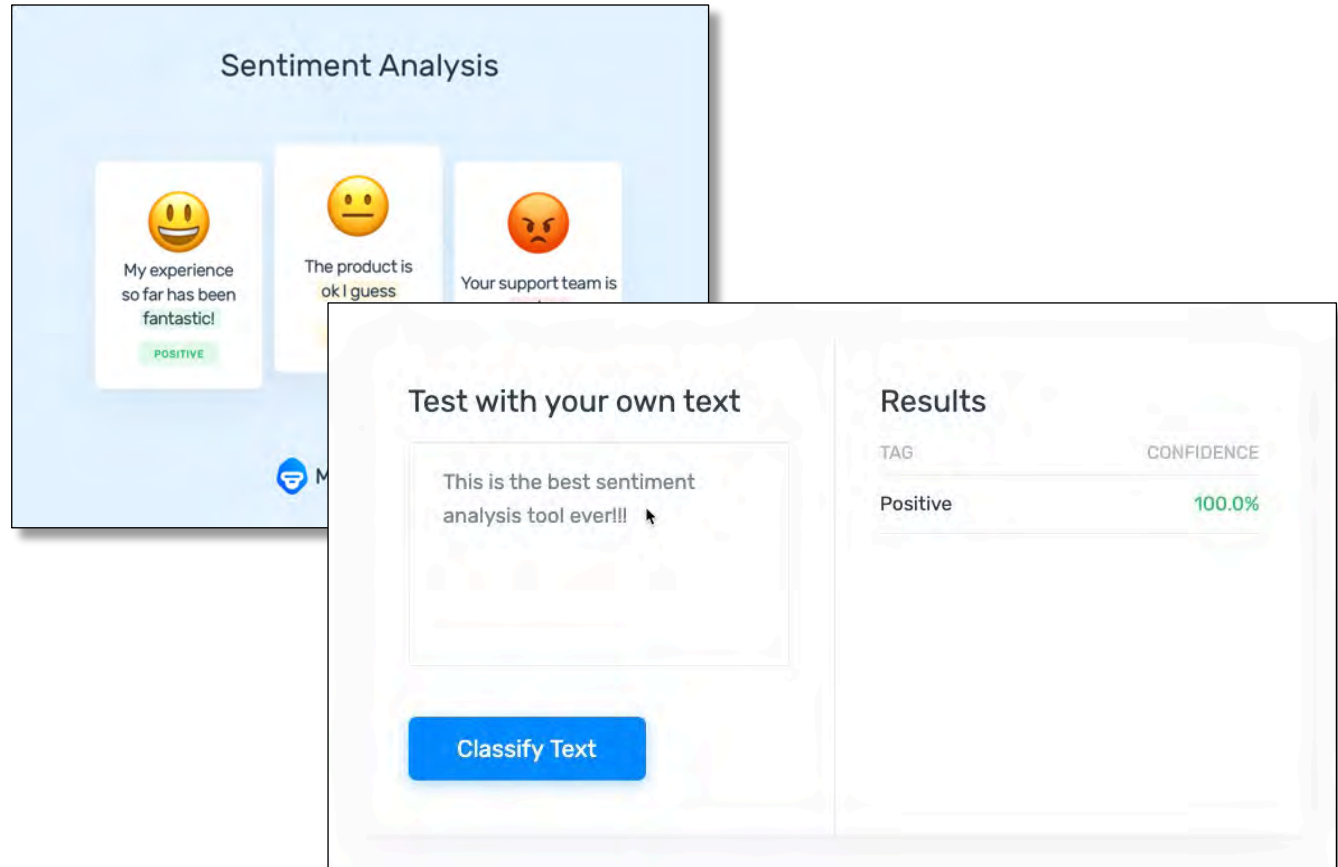
- Thrust network analysis (TNA)
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Thrust network analysis (TNA)

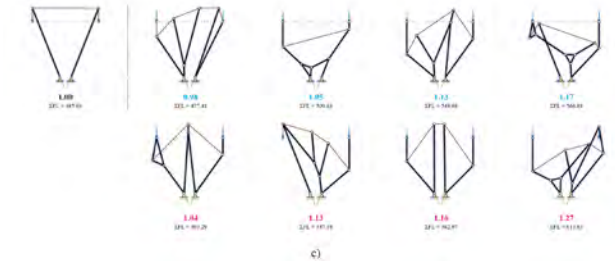
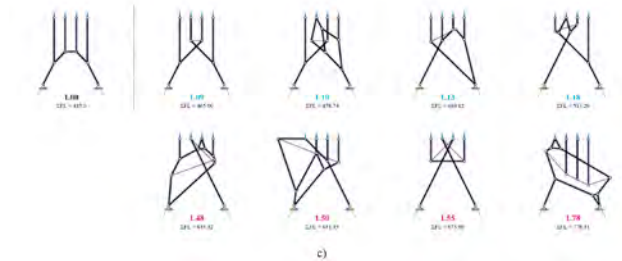
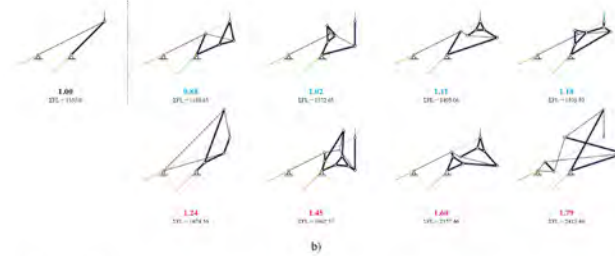
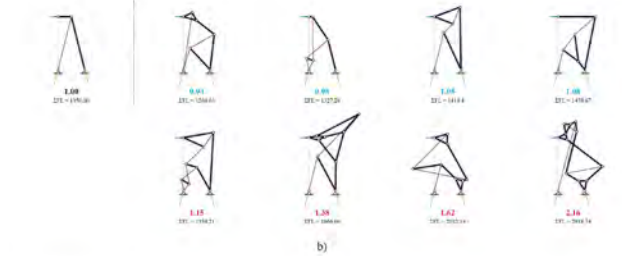
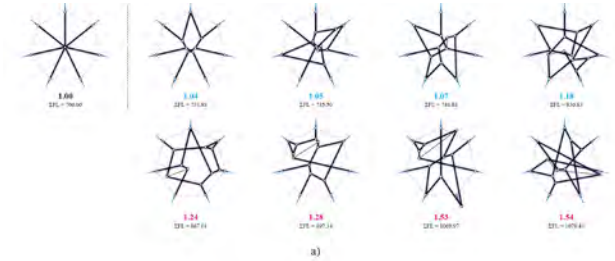
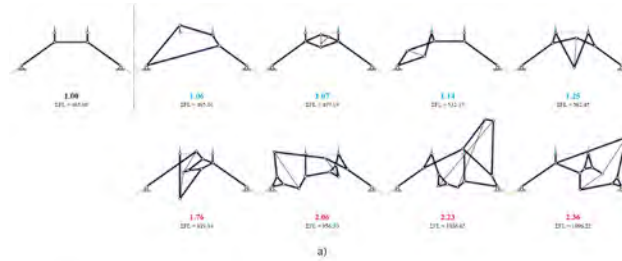
3D graphic statics

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[Shape grammars](#)

Visualisation



Form analysis

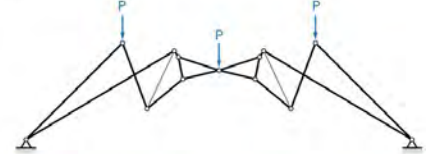
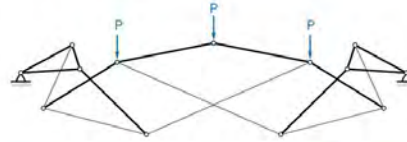
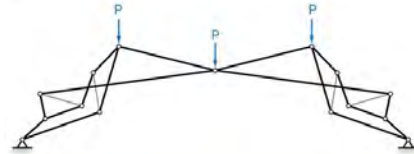
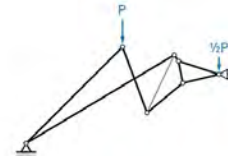
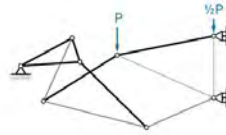
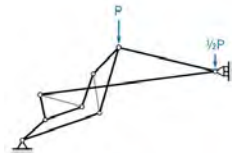
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Optimization



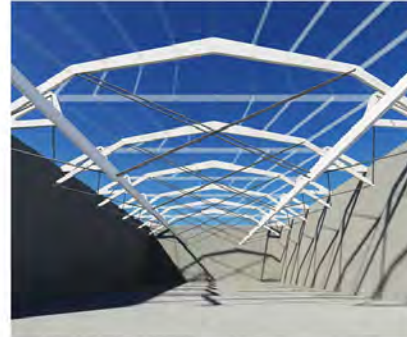
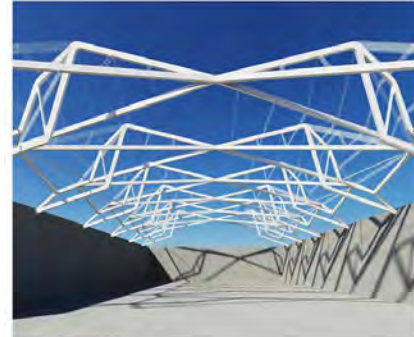
Form generation

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Visualisation

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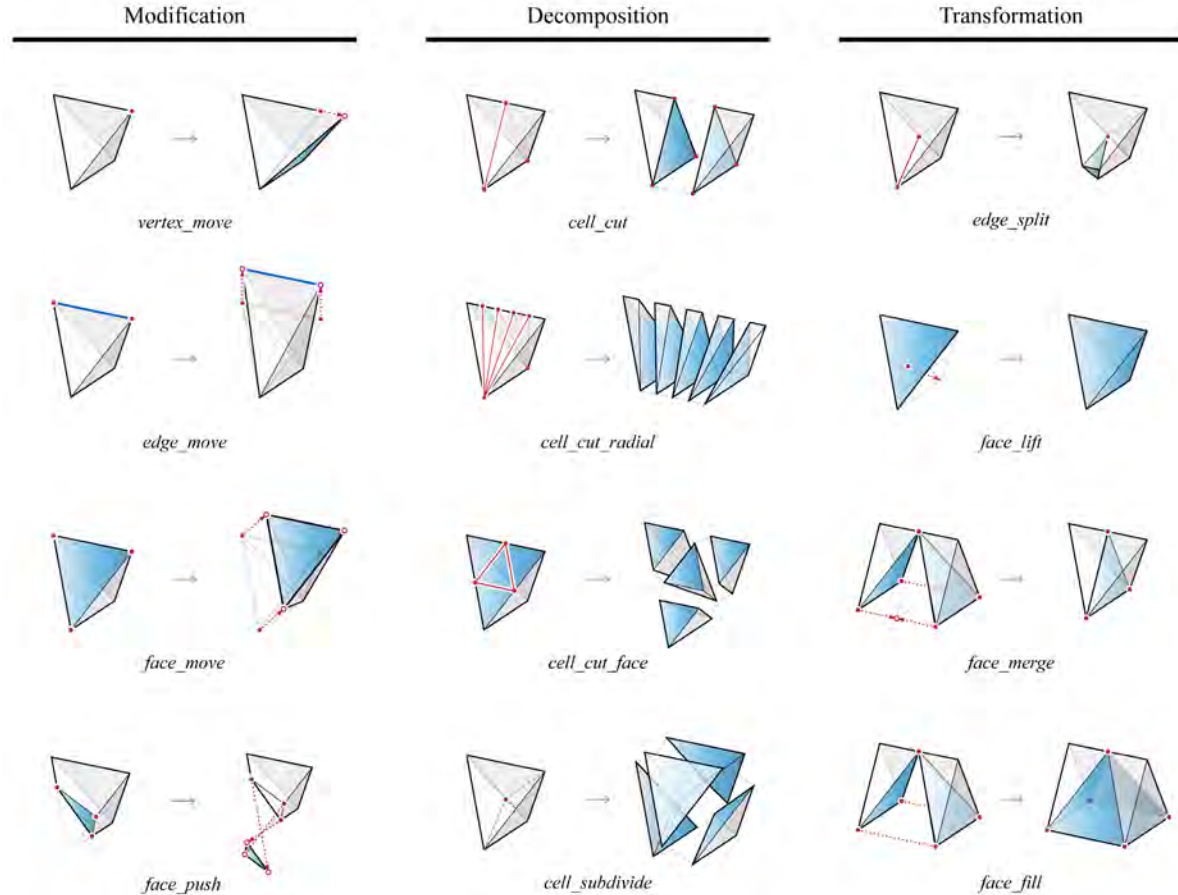
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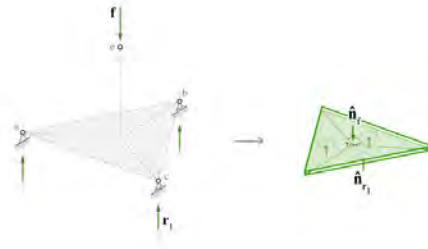
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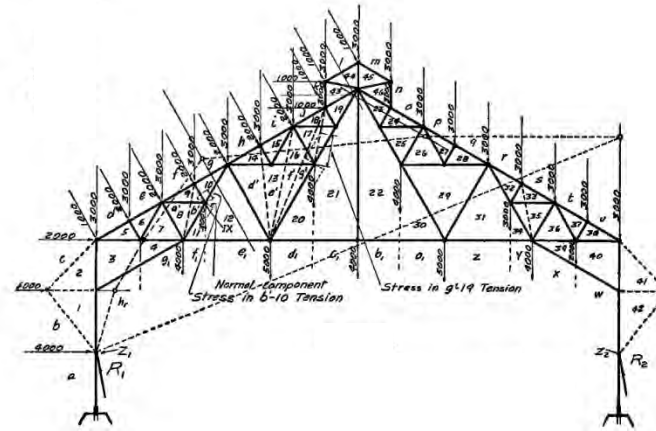
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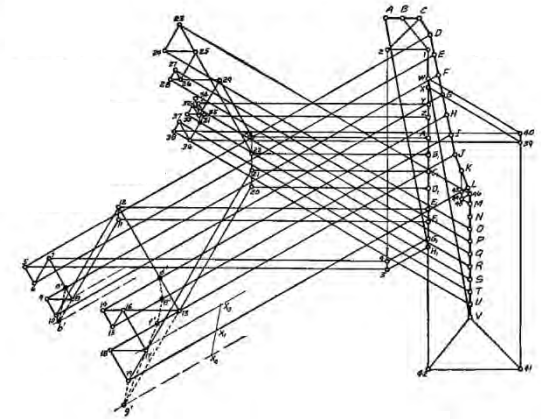
Generative design

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Visualisation



Form Diagram



Force Diagram

Form analysis

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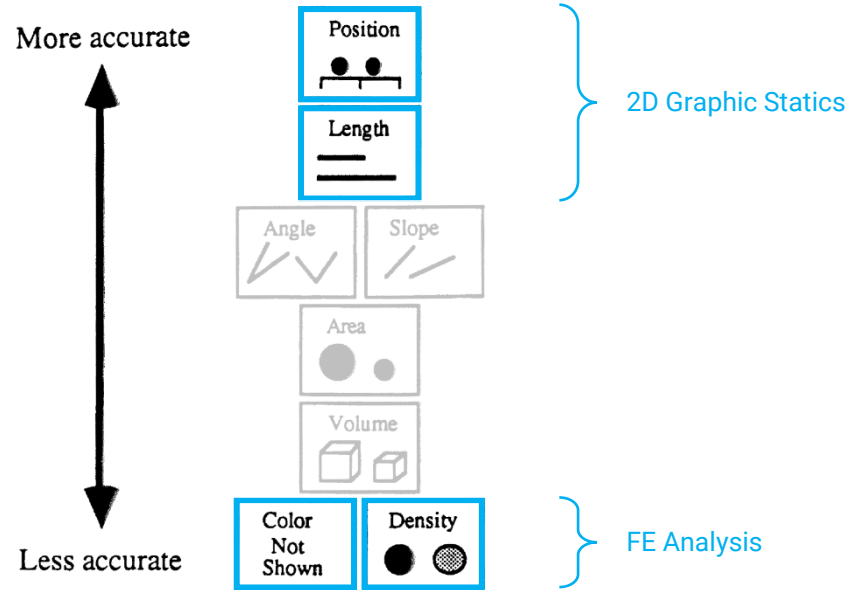
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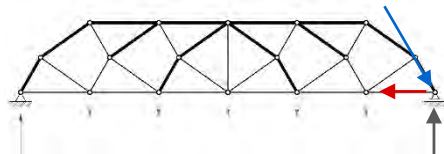
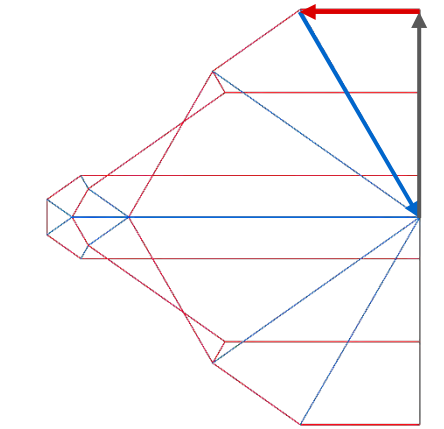
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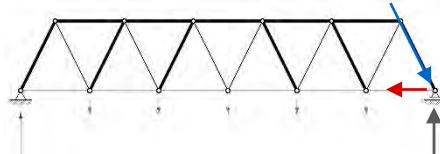
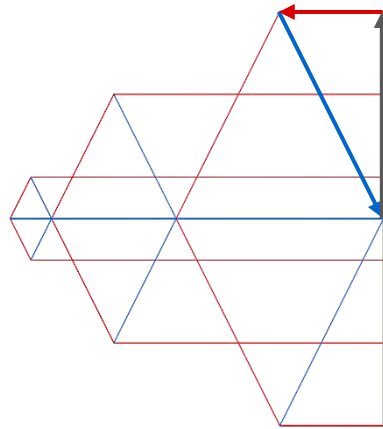


Accuracy and Effectiveness of Human Retinal Perception

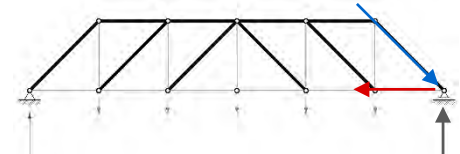
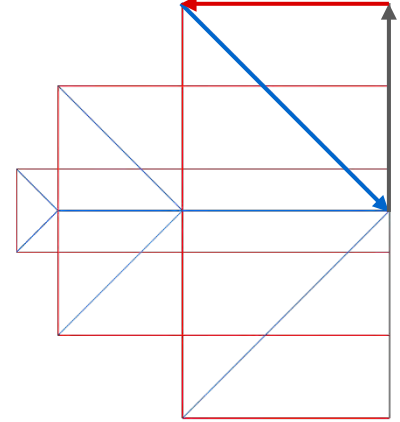
“Maxwell Diagram (perpendicular orientation)”



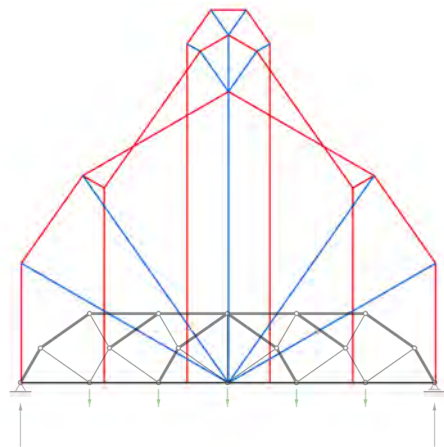
Michell



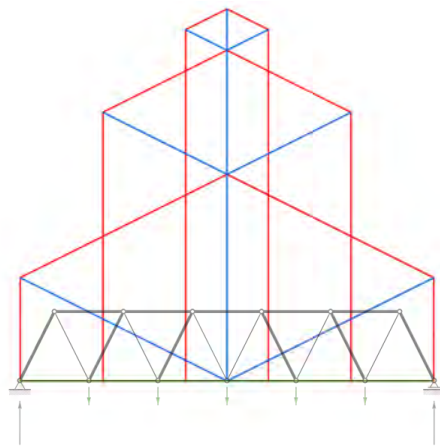
Warren



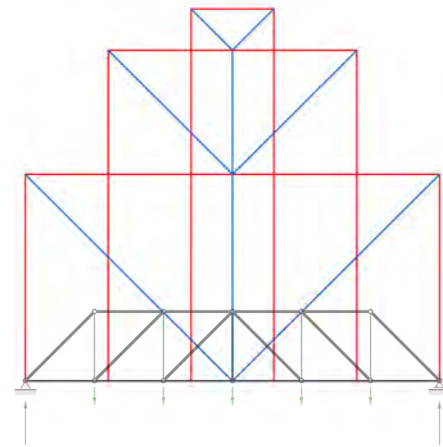
Howe



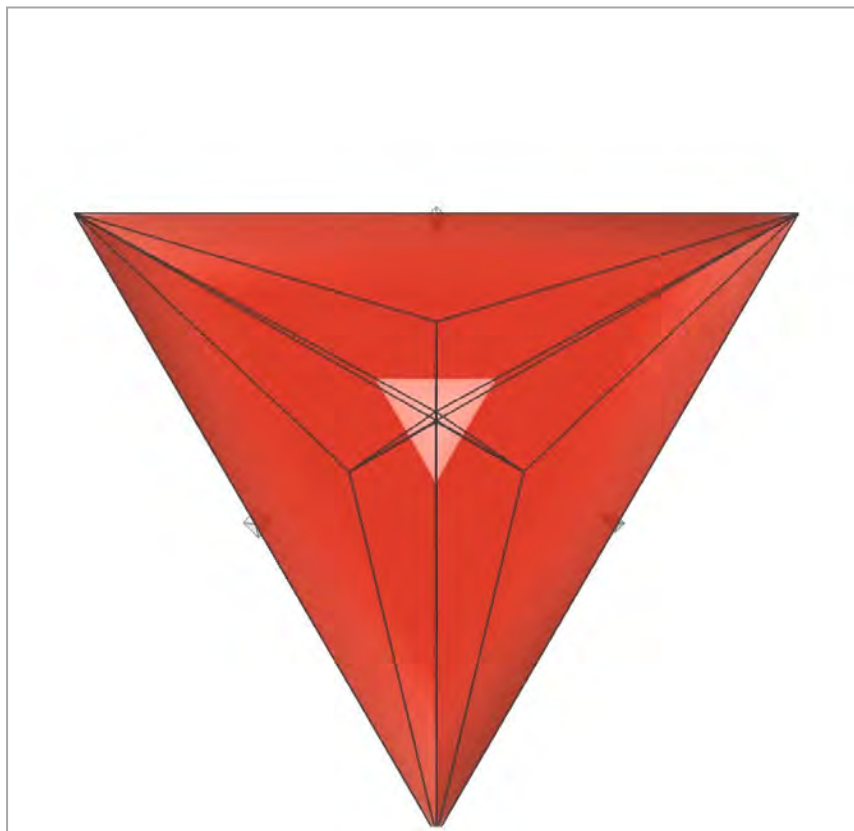
Michell



Warren

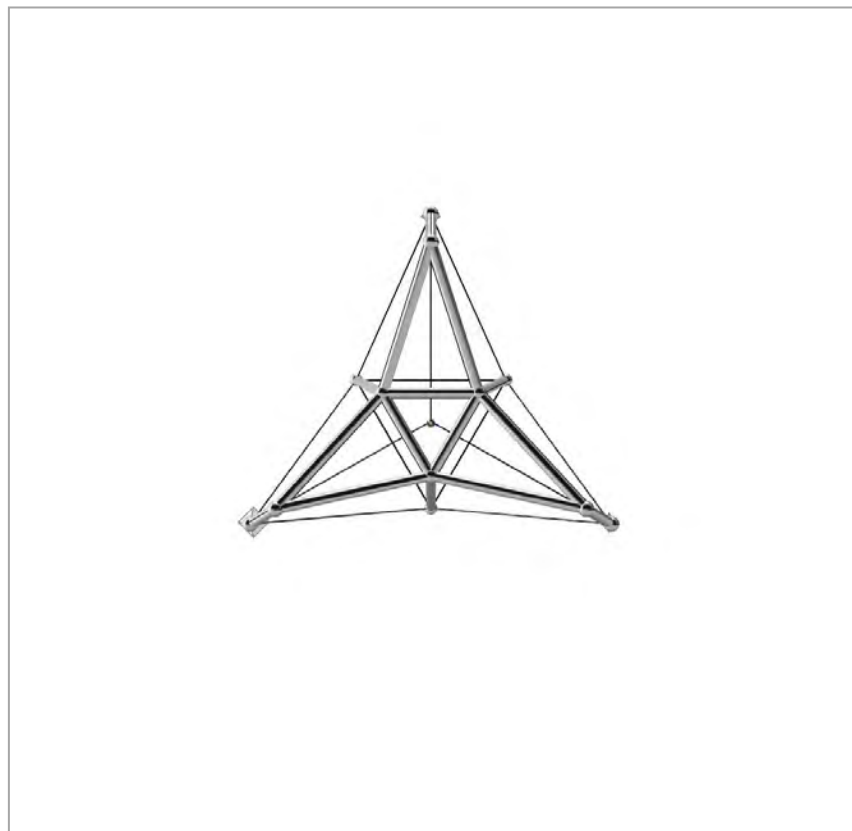


Howe

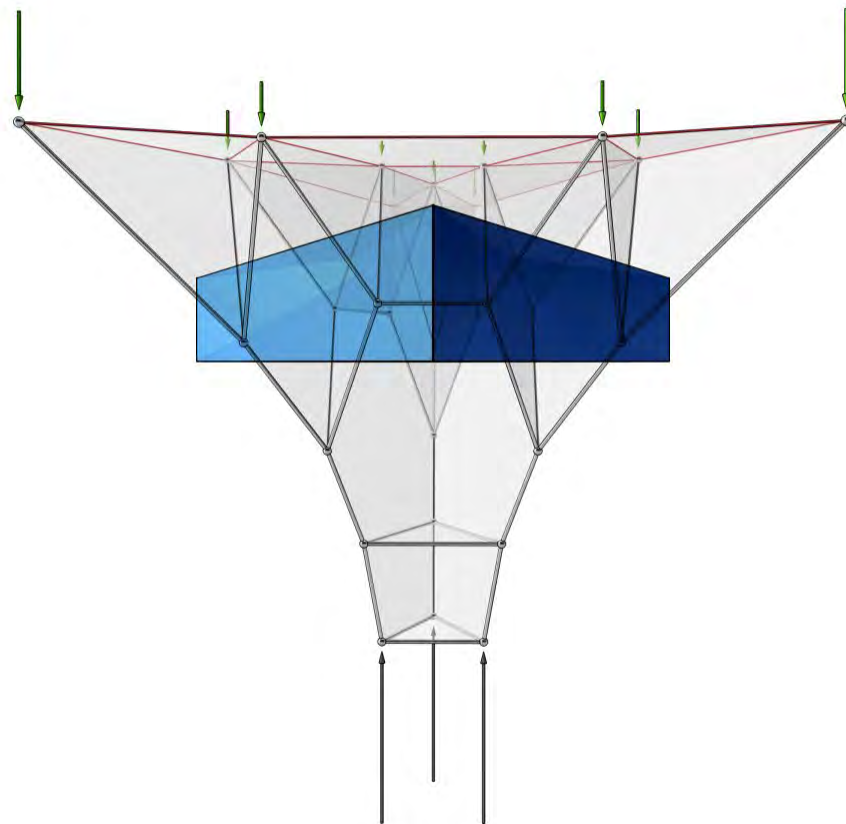


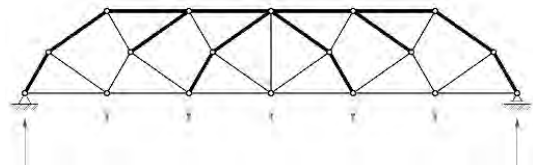
Interactive Polyhedral Force Diagram

← Synchronized Interface →

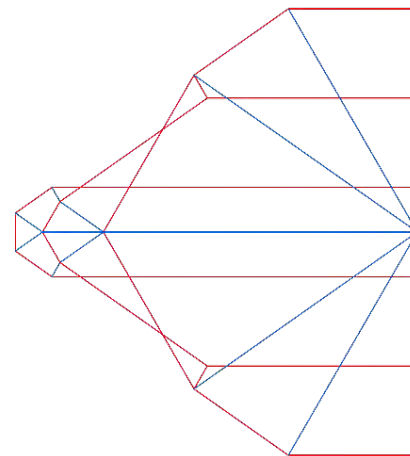


Form Diagram



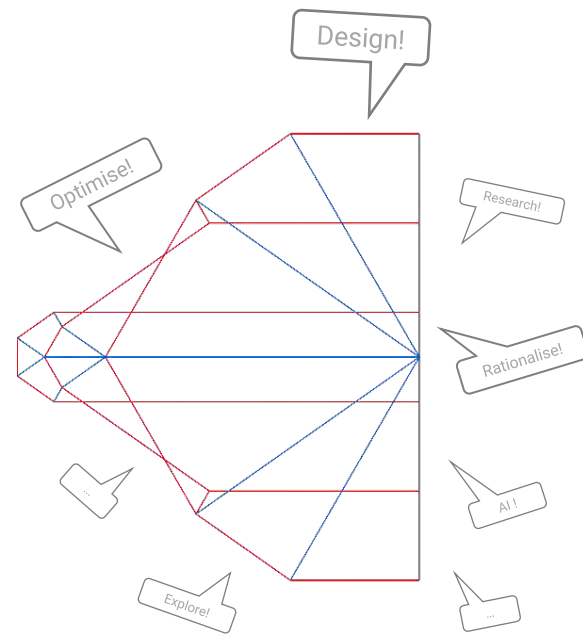


Geometry of form



Analyse

?



Geometry of forces

063-0605-00L : Computational Structural Design 1
Computational Graphic Statics

<http://block.arch.ethz.ch>



@blockresearchgroup

ETH zürich DARCH

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